Bayesian Quantile Deep Echo State Networks

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Abstract

Conditional quantiles are central to asymmetric decision losses, tail-risk assessment, and interval forecasting, but nonlinear dynamic quantile models can be difficult to fit when high-dimensional temporal features must be regularized while preserving uncertainty in quantile-specific readouts. We develop the quantile deep echo state network (Q–DESN), which uses a fixed deep echo state network feature map and places Bayesian inference only on the quantile readout. Conditional on reservoir construction, scaling, and washout, the readout is estimated with a quantile-fixed extended asymmetric Laplace working likelihood, ridge or regularized-horseshoe shrinkage priors, and either Markov chain Monte Carlo or a model-specific variational Bayes approximation with a Laplace–Delta treatment of the non-conjugate scale-asymmetry block. For multi-quantile reporting, we distinguish independent single-level fits followed by monotone synthesis from a joint quantile-vector readout that adapts Quantile-Varying Parameter shrinkage to adjacent quantile-specific readout coefficients. The joint prior encourages coherent quantile paths but does not by itself impose hard noncrossing constraints, so exact monotone output remains an explicit reporting contract. Synthetic studies, including a targeted interval-readout confirmation layer, and GloFAS streamflow and PriceFM electricity-price applications indicate that the resulting fixed-feature workflow is competitive under recorded protocols, while calibration, crossing behavior, and reservoir specification remain empirical diagnostics rather than automatic posterior guarantees.

1 Introduction

Forecast evaluation often requires summaries beyond the conditional mean. Conditional means are natural under squared-error loss, but asymmetric decision losses, tail-risk assessment, and interval forecasts call for conditional quantiles. Under asymmetric piecewise-linear loss, conditional quantiles are optimal point forecasts, and pairs of quantiles define interval forecasts (Koenker and Bassett, 1978; Koenker, 2005; Gneiting, 2011). Let y_{t+h} denote the future response at horizon h, and let $\mathcal{F}_t$ be the sigma-field generated by the information available at forecast origin t, including past responses and observed or scenario-specified covariates. With $F_{t,h}(u) = \Pr(y_{t+h} \le u \mid \mathcal{F}_t)$, define the level-$p_0$ conditional quantile by the generalized inverse

$$Q_{t,h}(p_0) = F_{t,h}^{-1}(p_0) = \inf\{u : F_{t,h}(u) \ge p_0\}, \qquad p_0 \in (0,1).$$

A single quantile level is sufficient when the decision target is fixed; a grid of fitted levels can be post-processed into an estimated predictive quantile function for interval construction or distributional forecast evaluation (Gneiting and Raftery, 2007; Gneiting and Katzfuss, 2014; Barlow and Brunk, 1972; Chernozhukov et al., 2010).

For nonlinear dynamic data, quantile forecast accuracy also depends on how temporal dependence is represented. Fixed reservoir features provide one computationally convenient representation: they encode recent history while leaving the recurrent weights outside the posterior inference problem. In the standard echo state network (ESN) construction used here, input and recurrent reservoir weights are generated once and then held fixed, so that only a readout is estimated (Jaeger, 2001; Lukoševičius and Jaeger, 2009; Lukoševičius, 2012). A deep echo state network (DESN) extends this idea by stacking reservoir layers, following the deep ESN construction of Gallicchio et al. (2018). Conditional on the selected architecture, random seed, scaling constants, reducers, and washout convention, the resulting reservoir states define a deterministic nonlinear feature matrix. The goal in this article is conditional quantile estimation and forecasting for nonlinear dynamic series without placing posterior uncertainty on the recurrent network: the DESN supplies the fixed nonlinear feature map, and Bayesian inference is concentrated in the readout model.

The quantile deep echo state network (Q–DESN) combines this fixed DESN design with a Bayesian quantile readout. For a chosen level p_0, the readout location $\mu_t = \boldsymbol{x}_t^\top \boldsymbol{\beta}$ is linked to the response through the quantile-fixed generalized asymmetric Laplace (GAL) distribution of Yan et al. (2025). Throughout this article, exAL denotes this quantile-fixed GAL working likelihood. Under this parameterization, μ_t is the model's p_0th conditional quantile, while the exAL asymmetry parameter allows mode, skewness, and tail behavior to vary relative to the asymmetric Laplace special case.

The exAL likelihood is used as a working likelihood in the Bayesian quantile-regression sense. It defines a posterior target for quantile-specific readout inference without asserting that exAL is the data-generating error law (Sriram et al., 2013; Yang et al., 2016; Ji et al., 2025). Posterior intervals for readout quantities and forecast bands generated from the fitted likelihood are therefore model-based summaries conditional on the fixed reservoir and the working likelihood. Predictive coverage is treated as an empirical diagnostic, assessed through simulation or validation data rather than assumed from the posterior alone.

High-dimensional readouts require regularization. The default coefficient prior is ridge, which gives dense Gaussian regularization. As an adaptive alternative, the regularized horseshoe introduces global-local shrinkage and a slab component for large coefficients (Carvalho et al., 2010; Piironen and Vehtari, 2017). Under the product representation used here, the local, global, and slab scale updates retain closed-form inverse-gamma forms (Nishimura and Suchard, 2023; Makalic and Schmidt, 2016). Posterior computation includes an MCMC sampler for the Q–DESN hierarchy. A mean-field variational Bayes approximation uses Laplace and Delta-method approximations for the non-conjugate exAL scale and asymmetry block. We refer to this implementation as VB–LD when the specific Laplace–Delta approximation needs to be distinguished. Its role is computational: it approximates the Q–DESN posterior under the working likelihood and is not presented as a general claim about variational-inference methodology.

The article contributes five elements. First, Q–DESN formulates Bayesian quantile readout inference conditional on fixed deep ESN features. Second, the readout combines a quantile-fixed exAL working likelihood with ridge regularization and a regularized-horseshoe adaptive-shrinkage alternative. Third, the MCMC and variational Bayes updates are derived for this fixed-design hierarchy. Fourth, multi-quantile reporting is treated through two distinct routes: independent single-level fits followed by monotone synthesis, and a joint quantile-vector readout that adapts QVP-style adjacent shrinkage to quantile-specific readout slopes. Fifth, a supplementary RQR–DESN generalized-Bayes interval-readout extension uses the fixed DESN feature map to estimate coverage-indexed intervals directly. Synthetic experiments and the GloFAS and PriceFM applications then illustrate the resulting workflow under reproducible forecast-validation protocols.

The remainder of the paper is organized as follows. Section 2 introduces notation, deep echo state network (DESN) dynamics, and the exAL representation. Section 3 presents the Q–DESN model and prior specification, including the joint quantile-vector extension. Section 4 describes posterior computation, while Sections 5–6 describe posterior prediction, optional quantile synthesis, and reservoir diagnostics. Section 7 presents the simulation design, including the RQR–DESN interval-readout confirmation, and Sections 8–9 give reproducible streamflow and electricity-price forecasting applications. Section 10 summarizes limitations and future work.

Related Work

Bayesian quantile regression provides the likelihood-based foundation for Q–DESN. Classical quantile regression targets conditional quantiles directly (Koenker and Bassett, 1978; Koenker, 2005). Bayesian implementations commonly use the asymmetric Laplace likelihood because of its connection to check-loss minimization and its mixture representation (Yu and Moyeed, 2001; Kozumi and Kobayashi, 2011). In this role, the asymmetric Laplace likelihood is a working likelihood: it defines a quantile-targeted posterior and supports conditional updates, while the coverage of model-based credible intervals may require separate assessment or adjustment under misspecification (Sriram et al., 2013; Yang et al., 2016; Ji et al., 2025). The quantile-fixed generalized asymmetric Laplace distribution of Yan et al. (2025) addresses a specific limitation of the asymmetric Laplace family by allowing the error distribution to vary after the target percentile has been fixed.

Dynamic quantile models provide statistical baselines. Dynamic quantile linear models combine dynamic linear modeling with Bayesian quantile regression and provide posterior computation for time-varying quantile paths (Gonçalves et al., 2020). The exDQLM of Barata et al. (2022) adds asymmetric likelihood flexibility for dynamic quantile inference. Q–DESN changes the predictor representation: instead of specifying a low-dimensional linear dynamic state, it conditions on fixed nonlinear reservoir states and places a Bayesian quantile readout on the resulting design matrix. The DQLM and exDQLM fits therefore serve as structured linear dynamic quantile baselines under the same article-level forecast protocol, rather than as a complete decomposition of representation, likelihood, and prior effects.

Reservoir computing and ESNs provide the nonlinear dynamic features used by the Q–DESN readout. ESNs were introduced by Jaeger (2001) and later reviewed within the broader reservoir-computing literature by Lukoševičius and Jaeger (2009); practical guidance is given by Lukoševičius (2012). Leaky-integrator ESNs make the leak rate an explicit time-scale parameter, while analyses of the echo state property caution that spectral-radius heuristics should be checked rather than treated as sufficient stability guarantees (Jaeger et al., 2007; Yildiz et al., 2012). Broader reservoir-computing surveys and time-series reviews discuss their role in forecasting and applied dynamic modeling (Zhang and Vargas, 2023; Yan et al., 2024; Cardoso et al., 2024; Sun et al., 2024). Deep ESNs, topology variants, and reservoir hyperparameter tuning extend the basic construction by stacking reservoirs, changing connectivity, or selecting reservoir parameters (Gallicchio et al., 2018; Kim and King, 2020; Maat et al., 2018; Bai et al., 2023; Viehweg et al., 2025). These sources motivate the fixed-feature viewpoint; Q–DESN places the Bayesian model on the quantile readout conditional on those features.

A related reservoir-computing literature studies uncertainty quantification and probabilistic forecasting. Deep ESNs with uncertainty quantification, probabilistic reservoir methods for load forecasting, activation-level uncertainty propagation, and Bayesian or state-space ESN formulations show that reservoir features can be embedded in probabilistic forecasting models (McDermott and Wikle, 2019; Guerra et al., 2023; Gandhi et al., 2018; Singh and Raman, 2025). This literature sets the boundary of the present contribution: uncertainty quantification for ESNs is already active. The narrower gap addressed here is quantile-specific Bayesian readout inference under a quantile-fixed asymmetric working likelihood with high-dimensional coefficient regularization.

Quantile-oriented ESN methods provide the closest forecasting comparison. Lv et al. (2016) use quantile-regression ESN ensembles to construct prediction intervals for blast-furnace gas generation, and Bonas et al. (2024) use penalized quantile regression to calibrate DESN forecasts for quasi-periodic climate processes. Q–DESN is complementary to these approaches. It does not treat the reservoir training or penalized-quantile calibration step as the Bayesian object; instead, conditional on the generated reservoir, it specifies an exAL readout whose posterior includes readout coefficients, likelihood scale and asymmetry, and shrinkage parameters. The joint extension further couples quantile-specific readout coefficients during inference, so raw multi-quantile paths can be made more coherent before any monotone reporting contract is applied.

Relaxed quantile regression (RQR) provides a different route to interval construction by learning interval roots directly rather than first estimating a preassigned lower and upper quantile pair (Pouplin et al., 2024). This avoids tying a target coverage level to an equal-tailed or symmetric quantile pair when the conditional error distribution may be skewed. We use this idea only as a supplementary fixed-feature interval readout: two DESN readout surfaces define ordered endpoints, and a generalized-Bayes update (Bissiri et al., 2016) targets the RQR loss. This extension is not a response likelihood and does not supply posterior predictive response draws; its evidence is therefore reported through interval score, empirical coverage, width, and baseline deltas.

Shrinkage priors enter because fixed reservoir designs can be high-dimensional and correlated. Ridge shrinkage provides the default dense Gaussian regularization. The regularized horseshoe, building on the horseshoe prior and its slab-regularized extension, is used as an adaptive global-local alternative (Carvalho et al., 2010; Piironen and Vehtari, 2017). Under the product representation used here, scale updates remain closed form (Nishimura and Suchard, 2023; Makalic and Schmidt, 2016). These priors regularize the readout; they should not be interpreted as guaranteeing feature-level interpretability without additional diagnostics.

Finally, a grid of quantile levels requires a distinction between post-processing and joint modeling. Separate Q–DESN fits can cross, and they do not learn dependence across quantile levels. Isotonic regression can project the fitted grid onto the monotone cone (Barlow and Brunk, 1972), and rearrangement methods provide a general approach to noncrossing quantile and probability curves (Chernozhukov et al., 2010). These operations repair the reported quantile curve after fitting; they do not define a joint posterior over quantile-indexed coefficients. Constrained quantile-regression methods instead impose non-crossing restrictions during estimation (Wu and Liu, 2009; Bondell et al., 2010). Recent work also shows that non-crossing constraints can be viewed as fused shrinkage across quantile-specific coefficients (Szendrei et al., 2025). Bayesian joint quantile models share information across levels through process priors, score-likelihood constructions, or smoothness penalties (Yang and Tokdar, 2017; Wu and Narisetty, 2021; Wang and Cai, 2024). The Quantile-Varying Parameter (QVP) construction of Kohns and Szendrei (2025) is especially relevant here because it places a state-space shrinkage prior on adjacent quantile-indexed coefficient changes. The joint Q–DESN extension below adapts that idea to fixed feature readouts and uses regularized-horseshoe shrinkage on adjacent slope innovations. The resulting prior is a soft non-crossing mechanism rather than a hard constraint. In the validation bundle below, it substantially reduces raw crossings relative to independent AL readouts before the monotone reporting contract is applied; exact monotonicity still requires an explicit ordered parameterization, projection, or rearrangement.

These distinctions determine the notation used below. We first separate the objects fixed by reservoir construction from the stochastic readout and likelihood quantities, so that posterior statements can be read as conditional on a recorded feature map.

2 Notation and Preliminaries

The notation separates deterministic reservoir quantities from stochastic readout and likelihood quantities. Table 1 summarizes the recurring objects. Matrices and vectors are bold, as in $\boldsymbol{W}$ and $\boldsymbol{x}$; scalar observations, latent states, and parameters use standard lowercase notation unless they are named constants. The symbols IG, GIG, and $\mathcal{N}^+$ denote inverse-gamma, generalized inverse Gaussian, and positive-truncated Normal distributions. For a square matrix $\boldsymbol{A}$, write $\rho(\boldsymbol{A}) = \max_i |\lambda_i(\boldsymbol{A})|$, where $\{\lambda_i(\boldsymbol{A})\}$ are the eigenvalues. Thus $\rho(\cdot)$ denotes spectral radius, while $\rho_d \in (0, 1)$ denotes the target spectral radius for layer d. Indices use $d = 1, \ldots, D$ and $t = 1, \ldots, T$. When conditioning information is shown explicitly, $Q_p(y \mid \cdot)$ denotes the conditional p-quantile of

Symbol	Description	Dimension and support
y_t	scalar response at time t	scalar
$p_0,\ p_k$	single-fit and synthesis-grid quantile levels	$(0,1)$
$\boldsymbol{z}_t$	exogenous covariates	$q \times 1$
$\boldsymbol{u}_t$	response lags and exogenous covariates	$(m+q) \times 1$
$\boldsymbol{h}_{t,d}$	reservoir state at layer d	$n_d \times 1$
$\tilde{\boldsymbol{h}}_{t,d}$	reduced state at layer d	$\tilde{n}_d \times 1$
α_d, ρ_d	layer leak and target spectral radius	$[0,1]$ and $(0,1)$
$\tilde{\boldsymbol{x}}_t$	feature vector without intercept	$r \times 1$
$\boldsymbol{x}_t$	readout design vector with intercept	$(r+1) \times 1$
$\boldsymbol{X}$	fixed DESN design matrix after washout	$T \times (r+1)$
$\boldsymbol{\beta}$	readout coefficients	$(r+1) \times 1$
κ_β^2	default ridge slope-prior variance	positive
σ, γ	exAL scale and asymmetry parameters	$\sigma > 0,\ \gamma \in (L, U)$
(λ_j, τ, ζ)	local, global, and slab shrinkage scales	positive

Table 1: Key notation used throughout. Reservoir quantities are drawn or constructed once, held fixed after washout, and collected in the design matrix $\boldsymbol{X}$.

the scalar response. The identity matrix, zero vector, and one vector of dimension m are denoted by $\boldsymbol{I}_m$, $\boldsymbol{0}_m$, and $\boldsymbol{1}_m$, respectively.

2.1 Deep Echo State Network (DESN)

The DESN is a deterministic feature map once the architecture, random seed, scaling constants, reducers, and washout convention are fixed. Conditional on these choices, posterior inference is restricted to the readout model. The main dimensions are stated before the recursion because the hierarchy contains several layer-specific matrices.

Notation and dimensions. The input vector is $\boldsymbol{u}_t = (y_{t-1}, \ldots, y_{t-m}, \boldsymbol{z}_t^\top)^\top \in \mathbb{R}^{m+q}$, with m response lags and q exogenous variables. For layer d, $\boldsymbol{W}_d \in \mathbb{R}^{n_d \times n_d}$ is recurrent, $\boldsymbol{W}_1^{\text{in}} \in \mathbb{R}^{n_1 \times (m+q)}$ maps the original input to the first layer, and $\boldsymbol{W}_d^{\text{in}} \in \mathbb{R}^{n_d \times \tilde{n}_{d-1}}$ maps the reduced state from layer $d-1$ to layer d for $d \geq 2$. The reducer $\boldsymbol{Q}_{d-1} \in \mathbb{R}^{\tilde{n}_{d-1} \times n_{d-1}}$ maps $\boldsymbol{h}_{t,d-1} \in \mathbb{R}^{n_{d-1}}$ to $\tilde{\boldsymbol{h}}_{t,d-1} \in \mathbb{R}^{\tilde{n}_{d-1}}$. The readout dimension is $r = n_D + \sum_{d=1}^{D-1} \tilde{n}_d + (m+q)$, and $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_r)^\top \in \mathbb{R}^{r+1}$.

Random sparse weights and spectral scaling. In contrast to fully trained recurrent networks, reservoir and input weights are drawn once and kept fixed throughout inference. Let $\circ$ denote the Hadamard product. Each recurrent matrix is $\boldsymbol{W}_d = \Phi_d^{(W)} \circ Z_d^{(W)}$, and each input matrix is generated analogously as $\boldsymbol{W}_d^{\text{in}} = \Phi_d^{(\text{in})} \circ Z_d^{(\text{in})}$. Mask entries are Bernoulli with probabilities $\pi_d^{(W)}$ and $\pi_d^{(\text{in})}$; base entries are independent draws from zero-mean finite-variance distributions p_W and p_{in}, such as $\mathcal{N}(0, \sigma^2)$ or $\text{Unif}[-a, a]$. Masks and base draws are mutually independent and independent across entries and layers. The recurrent matrices are then scaled to the target spectral radii,

$$\text{Spectral-radius scaling:} \qquad \bar{\boldsymbol{W}}_d \;=\; \frac{\rho_d}{\rho(\boldsymbol{W}_d)}\, \boldsymbol{W}_d, \qquad \rho_d \in (0,1).$$

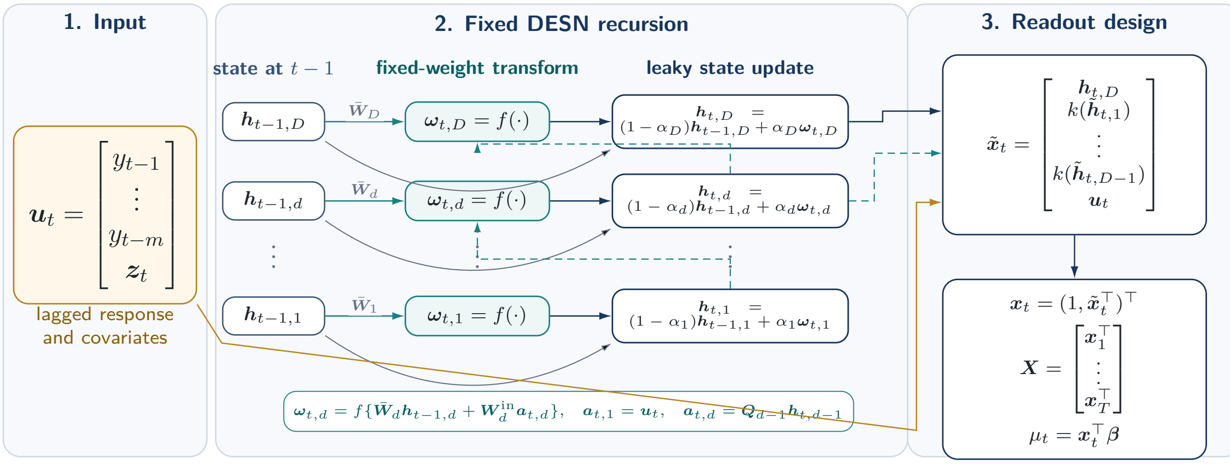

Figure 1: Fixed DESN feature construction. The input block collects response lags and exogenous covariates. The reservoir block shows that each layer update combines the previous state with a nonlinear innovation driven by fixed recurrent weights and, for upper layers, the current-time reduced state from the layer below. After washout, the top-layer state, transformed lower-layer reductions, and retained input block form the fixed readout design used by the Q–DESN likelihood.

If $\rho(\boldsymbol{W}_d) = 0$, the matrix is resampled. Input matrices $\boldsymbol{W}_d^{\text{in}}$ are not spectrally scaled.

DESN recursion and readout features. Figure 1 summarizes the three objects carried by the fixed feature construction: the lagged input, the layerwise reservoir update, and the readout design.

With layer-specific leak rates $\alpha_d \in [0, 1]$, elementwise nonlinearity $f(\cdot)$, and lower-layer transformation $k(\cdot)$, the fixed feature map is

$$
\begin{aligned}
\text{Layer 1 innovation}: &\quad \boldsymbol{\omega}_{t,1} = f\left(\bar{\boldsymbol{W}}_1 \boldsymbol{h}_{t-1,1} + \boldsymbol{W}_1^{\text{in}} \boldsymbol{u}_t\right), \\
\text{Layer 1 update}: &\quad \boldsymbol{h}_{t,1} = (1-\alpha_1)\boldsymbol{h}_{t-1,1} + \alpha_1 \boldsymbol{\omega}_{t,1}, \\
\text{Reduced bridge state}: &\quad \tilde{\boldsymbol{h}}_{t,d-1} = \boldsymbol{Q}_{d-1} \boldsymbol{h}_{t,d-1}, && d = 2, \ldots, D, \\
\text{Layer d innovation}: &\quad \boldsymbol{\omega}_{t,d} = f\left(\bar{\boldsymbol{W}}_d \boldsymbol{h}_{t-1,d} + \boldsymbol{W}_d^{\text{in}} \tilde{\boldsymbol{h}}_{t,d-1}\right), && d = 2, \ldots, D, \\
\text{Layer d update}: &\quad \boldsymbol{h}_{t,d} = (1-\alpha_d)\boldsymbol{h}_{t-1,d} + \alpha_d \boldsymbol{\omega}_{t,d}, && d = 2, \ldots, D, \\
\text{Readout feature stack}: &\quad \tilde{\boldsymbol{x}}_t = \left[\boldsymbol{h}_{t,D}^\top,\ k(\tilde{\boldsymbol{h}}_{t,1})^\top, \ldots, k(\tilde{\boldsymbol{h}}_{t,D-1})^\top,\ \boldsymbol{u}_t^\top\right]^\top, \\
\text{Intercept and readout}: &\quad \boldsymbol{x}_t = (1, \tilde{\boldsymbol{x}}_t^\top)^\top, \qquad \mu_t = \boldsymbol{x}_t^\top \boldsymbol{\beta}.
\end{aligned}
\tag{2.1}
$$

Here $k(\cdot)$ is an elementwise activation applied to reduced lower-layer states, commonly $k = \text{id}$ or $k = f$.

The input vector $\boldsymbol{u}_t$ bundles response lags and exogenous covariates; the intercept is appended only in $\boldsymbol{x}_t$. The leaky update combines persistence $(1-\alpha_d)\boldsymbol{h}_{t-1,d}$ with the nonlinear innovation $\boldsymbol{\omega}_{t,d}$. The leak rate α_d and target radius ρ_d control the empirical time scale and persistence of the deterministic state recursion; they are tuning choices rather than sufficient guarantees of the echo state property. Reducers $\boldsymbol{Q}_{d-1}$ may be fixed random projections, optionally orthonormalized, or precomputed linear reducers such as PCA from an initial preprocessing pass.

The feature stack concatenates the top-layer state, transformed reduced lower-layer states, and $\boldsymbol{u}_t$, yielding $\tilde{\boldsymbol{x}}_t \in \mathbb{R}^r$. Appending an intercept gives $\boldsymbol{x}_t \in \mathbb{R}^{r+1}$, and the readout maps this feature vector to the scalar location μ_t. States are initialized at $\boldsymbol{h}_{0,d} = \boldsymbol{0}_{n_d}$, or from $\mathcal{N}(\boldsymbol{0}_{n_d}, \sigma^2_{h,d}\boldsymbol{I}_{n_d})$. A teacher-forcing washout $t = 1, \ldots, T_0$ is run before features enter the readout. Washout length and scaling choices can affect short-horizon behavior. All masks and base matrices are fixed for the entire analysis and are not resampled during posterior inference.

2.2 Extended Asymmetric Laplace (exAL)

The symbol exAL denotes the quantile-fixed generalized asymmetric Laplace (GAL) distribution of Yan et al. (2025). We use the name exAL to emphasize that it extends the asymmetric Laplace (AL) working likelihood while keeping the target quantile fixed. The standard AL likelihood is recovered as a special case, but its skewness is fixed once p_0 is chosen. The exAL asymmetry parameter allows mode, skewness, and tail behavior to vary within a quantile-fixed likelihood. At fixed level p_0, μ is the p_0th quantile, $\sigma > 0$ is a scale parameter, and γ is an asymmetry parameter with bounded support (L, U). The endpoints L and U solve $g(\gamma) = 1 - p_0$ and $g(\gamma) = p_0$, respectively, where $g(\gamma) = 2\Phi(-|\gamma|)\exp(\gamma^2/2)$, and Φ is the standard Normal CDF.

Posterior computation uses the stochastic representation of the exAL distribution. If

$$y \sim \mathrm{exAL}_{p_0}(\mu, \sigma, \gamma),$$

then there exist independent $\epsilon \sim \mathcal{N}(0,1)$, $z \sim \mathrm{Exp}(1)$, and $s \sim \mathcal{N}^+(0,1)$ such that

$$y = \mu + \lambda(\gamma)\sigma s + A(\gamma)\sigma z + \{B(\gamma)\sigma^2 z\}^{1/2}\epsilon,$$

where $A(\gamma) = \{1 - 2p_\gamma\}/\{p_\gamma(1-p_\gamma)\}$, $B(\gamma) = 2/\{p_\gamma(1-p_\gamma)\}$, $C(\gamma) = \{\mathbf{1}\{\gamma > 0\} - p_\gamma\}^{-1}$, $p_\gamma = \mathbf{1}\{\gamma < 0\} + \{p_0 - \mathbf{1}\{\gamma < 0\}\}/g(\gamma)$, and $\lambda(\gamma) = C(\gamma)|\gamma|$. Setting $v = \sigma z$ gives $v \mid \sigma \sim \mathrm{Exp}(\mathrm{rate} = 1/\sigma)$ and the conditional Gaussian form used for Q–DESN posterior computation:

$$y \mid \mu, \sigma, \gamma, s, v \sim \mathcal{N}(\mu + \lambda(\gamma)\sigma s + A(\gamma)v,\ B(\gamma)\sigma v).$$

All quantities $g(\gamma)$, p_γ, $A(\gamma)$, $B(\gamma)$, $C(\gamma)$, and $\lambda(\gamma)$ depend on the target quantile level p_0; this dependence is suppressed for readability. Once (p_0, γ) is fixed, these are deterministic constants in the Gaussian augmentation.

3 Model Specification

Fix a quantile level $p_0 \in (0,1)$. After reservoir construction and washout, $\boldsymbol{X} = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T)^\top$ is treated as fixed, and all posterior uncertainty is conditional on this design. The statistical model is a Bayesian quantile readout for $\boldsymbol{y}$ given $\boldsymbol{X}$. The observation model and priors define the posterior target; the Gaussian augmentation below is a computational representation of the asymmetric working likelihood. Dependence on p_0 is suppressed when no ambiguity arises.

3.1 Quantile Deep Echo State Network (Q–DESN)

The Q–DESN is obtained by inserting the fixed DESN feature map into the quantile-fixed exAL likelihood of Section 2.2. Let $\tilde{\boldsymbol{x}}_t$ be the feature vector in Eq. (2.1), and set $\boldsymbol{x}_t = (1, \tilde{\boldsymbol{x}}_t^\top)^\top$. Conditional on the fixed design matrix, the marginal working likelihood is

$$y_t \mid \boldsymbol{x}_t, \boldsymbol{\beta}, \sigma, \gamma \sim \mathrm{exAL}_{p_0}(\boldsymbol{x}_t^\top \boldsymbol{\beta}, \sigma, \gamma), \qquad t = 1, \dots, T.$$

Thus the dynamic readout $\boldsymbol{x}_t^\top \boldsymbol{\beta}$ is the model's conditional p_0-quantile, while σ and γ control the scale and shape of the working error distribution. For posterior computation, the same exAL mixture representation is applied at each time point:

$$\begin{aligned}
\text{Quantile readout :}& \quad \mu_t = \boldsymbol{x}_t^\top \boldsymbol{\beta}, \qquad \boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_r)^\top \in \mathbb{R}^{r+1}, \\
\text{Gaussian conditional :}& \quad y_t \mid \boldsymbol{\beta}, \sigma, \gamma, s_t, v_t, \boldsymbol{x}_t \sim \mathcal{N}\Big(\mu_t + \lambda(\gamma)\sigma s_t + A(\gamma) v_t,\, B(\gamma)\sigma v_t\Big), \\
\text{Positive-shift latent :}& \quad s_t \sim \mathcal{N}^+(0, 1), \\
\text{Scale-mixture latent :}& \quad v_t \mid \sigma \sim \mathrm{Exp}(\text{rate} = 1/\sigma).
\end{aligned} \tag{3.1}$$

The display separates the statistical readout from the computational augmentation. The first line defines the quantile path; the second gives the conditional Gaussian observation equation; and the last two lines introduce auxiliary variables. The constants $A(\gamma)$, $B(\gamma)$, and $\lambda(\gamma)$ are inherited from Section 2.2. The DESN feature map is fixed before posterior inference, while $\boldsymbol{\beta}$, $\sigma > 0$, and $\gamma \in (L, U)$ are posterior unknowns. Integrating out (s_t, v_t) recovers the marginal exAL likelihood, so these variables do not alter the quantile target; they expose conditionally Gaussian updates. Complete-data joint distributions for the AL and exAL likelihoods with ridge and RHS priors are given in Section S4 of the supplement.

3.2 Single-Level Prior Specification

The intercept is modeled separately from the slope coefficients: $\beta_0 \sim \mathcal{N}(b_0, V_0)$, with $b_0 \in \mathbb{R}$ and $V_0 > 0$. This separation avoids forcing the baseline quantile level toward zero when the readout features are not centered around the target quantile.

For the non-intercept coefficients $\boldsymbol{\beta}_{1:r} = (\beta_1, \dots, \beta_r)^\top$, the default ridge prior is

$$\boldsymbol{\beta}_{1:r} \mid \kappa_\beta^2 \sim \mathcal{N}(\boldsymbol{0}_r,\, \kappa_\beta^2 \boldsymbol{I}_r), \tag{3.2}$$

where $\kappa_\beta^2 > 0$ is fixed or assigned a weakly informative scale prior. This Gaussian prior gives dense baseline regularization.

The regularized horseshoe (RHS) prior is the adaptive shrinkage alternative. It starts from the ordinary horseshoe (Carvalho et al., 2010), adds slab regularization (Piironen and Vehtari, 2017), and uses the product representation of Nishimura and Suchard (2023) to retain closed-form scale updates. For non-intercept coefficients $j = 1, \ldots, r$, the ordinary horseshoe is

$$\beta_j \mid \tau, \lambda_j \sim \mathcal{N}(0, \tau^2\lambda_j^2), \qquad \lambda_j \sim C^+(0,1), \qquad \tau \sim C^+(0, \tau_0), \tag{3.3}$$

where τ is the global scale and λ_j is coefficient-specific. This prior concentrates mass near zero while allowing some coefficients to escape strong shrinkage, but it does not separately control the largest prior variances. The regularized version introduces a slab scale $\zeta > 0$ and sets

$$V_j(\lambda_j, \tau, \zeta) = \left(\zeta^{-2} + \tau^{-2}\lambda_j^{-2}\right)^{-1} = \frac{\zeta^2\tau^2\lambda_j^2}{\zeta^2 + \tau^2\lambda_j^2}, \qquad j = 1, \ldots, r, \tag{3.4}$$

with conditional prior

$$\beta_j \mid \lambda_j, \tau, \zeta \sim \mathcal{N}(0, V_j), \qquad j = 1, \ldots, r. \tag{3.5}$$

This is the RHS coefficient prior used in the adaptive-shrinkage Q–DESN fits. Equation (3.4) gives the interpretation: if $\tau^2\lambda_j^2 \ll \zeta^2$, then $V_j \approx \tau^2\lambda_j^2$, recovering ordinary horseshoe behavior; if $\tau^2\lambda_j^2 \gg \zeta^2$, then $V_j \approx \zeta^2$, so the slab caps the effective prior variance.

The direct Piironen–Vehtari joint formulation gives the desired regularized horseshoe coefficient law. For Gibbs and CAVI updates, however, its global-local block includes an extra $\{1+\tau^2\lambda_j^2/\zeta^2\}^{1/2}$ factor, so the ordinary horseshoe updates for (τ, λ_j) are not preserved. The Nishimura–Suchard joint construction preserves the same conditional coefficient law while restoring closed-form scale updates:

$$p(\beta_j, \lambda_j \mid \tau, \zeta) \propto \exp\left(-\frac{\beta_j^2}{2\tau^2\lambda_j^2}\right) \exp\left(-\frac{\beta_j^2}{2\zeta^2}\right) \lambda_j^{-1}(1+\lambda_j^2)^{-1}, \tag{3.6}$$

where proportionality is with respect to (β_j, λ_j). The (τ, ζ)-dependent Gaussian normalizing factors are retained in the full-conditionals derivations so that the inverse-gamma shape parameters are recovered correctly. Thus the RHS option adds adaptive global-local shrinkage without losing inverse-gamma updates for the local and global scales.

Closed-form scale updates use the inverse-gamma representation of half-Cauchy priors (Makalic and Schmidt, 2016):

$$\lambda_j^2 \mid \nu_j \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\nu_j}\right), \qquad \nu_j \sim \mathrm{IG}\left(\frac{1}{2}, 1\right), \quad j = 1, \ldots, r, \tag{3.7}$$

$$\tau^2 \mid \xi \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\xi}\right), \qquad \xi \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\tau_0^2}\right). \tag{3.8}$$

The slab scale is either fixed or assigned

$$\zeta^2 \sim \mathrm{IG}(a_\zeta, b_\zeta). \tag{3.9}$$

Under (3.6), this prior remains compatible with closed-form updating (Nishimura and Suchard, 2023).

The likelihood parameters are assigned $\sigma \sim \mathrm{IG}(a_\sigma, b_\sigma)$ and $\gamma \sim \pi_\gamma(\gamma)\,\mathbf{1}\{L < \gamma < U\}$, with (L, U) determined by the quantile-fixing construction as described in Section 2.2. For RHS fits, collect $\Theta = (\boldsymbol{\beta}, \sigma, \gamma, \boldsymbol{\lambda}^2, \tau^2, \boldsymbol{\nu}, \xi, \zeta^2)$, where $\boldsymbol{\lambda}^2 = (\lambda_1^2, \ldots, \lambda_r^2)^\top$ and $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_r)^\top$. The entry ζ^2 is omitted when the slab scale is fixed. Ridge fits replace the global-local shrinkage block by κ_β^2 when that scale is assigned a prior.

3.3 Joint Quantile-Vector Q–DESN

The single-level model can be fit independently over a grid of quantile levels, but independent fitting does not couple readout coefficients across levels. For applications in which the coefficient path itself is part of the inferential target, let $0 < p_1 < \cdots < p_Q < 1$ be a fixed quantile grid and write $\boldsymbol{Z}$ for the $T \times r$ non-intercept fixed feature design. In a full Q–DESN analysis, $\boldsymbol{Z}$ is the non-intercept DESN design. The joint synthetic validation below uses the same readout on a frozen article-facing feature design recorded by that validation bundle; it does not change the model definition. The joint quantile-vector readout separates intercepts from high-dimensional slopes:

$$Q_{p_q}(y_t \mid \mathcal{F}_t) = \alpha_q + \tilde{\boldsymbol{x}}_t^\top \boldsymbol{\beta}_q, \qquad q = 1, \ldots, Q, \quad t = 1, \ldots, T, \tag{3.10}$$

where $\alpha_q \in \mathbb{R}$ is the quantile-specific intercept and $\boldsymbol{\beta}_q \in \mathbb{R}^r$ is the slope readout at level p_q. The intercepts are not included in the high-dimensional RHS slope path by default; they are either weakly regularized independently or represented through ordered gaps when a stronger monotonicity convention is desired.

This construction generalizes the single-level RHS Q–DESN by replacing the single intercept-slope readout $(\beta_0, \boldsymbol{\beta}_{1:r})$ with $\{(\alpha_q, \boldsymbol{\beta}_q)\}_{q=1}^{Q}$. For $Q = 1$, the individual RHS Q–DESN is recovered by taking α_1 as the single-level intercept and identifying $\boldsymbol{\beta}_1$ with the single-level non-intercept slope vector, with no learned slope baseline. Equivalently, set $\boldsymbol{\beta}_{\mathrm{base}} \equiv \mathbf{0}_r$ and place the RHS prior directly on $\Delta_1 = \boldsymbol{\beta}_1$.

For each q, the working likelihood uses the quantile-fixed exAL family at the corresponding target level,

$$y_t \mid \alpha_q, \boldsymbol{\beta}_q, \sigma_q, \gamma_q \sim \mathrm{exAL}_{p_q}\Big(\alpha_q + \tilde{\boldsymbol{x}}_t^\top \boldsymbol{\beta}_q, \sigma_q, \gamma_q\Big). \tag{3.11}$$

The Q likelihood contributions form a composite working likelihood over the same response series. A learning-rate parameter $\kappa > 0$ may be used to temper this composite contribution, with $\kappa = 1$ as the default and $\kappa = 1/Q$ as a conservative sensitivity setting under the general Bayes view of loss-based updating (Bissiri et al., 2016). This composite target updates beliefs about the quantile-indexed readout path; it is not used as a normalized data-generating density for a future response.

For a genuine multi-quantile grid, the joint prior adapts the Quantile-Varying Parameter (QVP) idea that adjacent quantile-indexed coefficient changes should be small unless the data support tail-specific effects (Kohns and Szendrei, 2025). Let $\boldsymbol{\beta}_{\mathrm{base}} \in \mathbb{R}^r$ be a baseline readout and define adjacent innovations

$$\Delta_{1,j} = \beta_{1,j} - \beta_{\mathrm{base},j}, \qquad \Delta_{q,j} = \beta_{q,j} - \beta_{q-1,j}, \quad q = 2, \ldots, Q.$$

The baseline slope receives an ordinary RHS prior, while the adjacent innovations receive RHS shrinkage with quantile-gap-specific global scales:

$$\beta_{\mathrm{base},j} \mid \lambda_j^0, \tau^0, \zeta_0 \sim \mathcal{N}(0, V_j^0), \qquad V_j^0 = \frac{\zeta_0^2 (\tau^0)^2 (\lambda_j^0)^2}{\zeta_0^2 + (\tau^0)^2 (\lambda_j^0)^2}, \tag{3.12}$$

$$\Delta_{q,j} \mid \lambda_{q,j}^\Delta, \tau_q^\Delta, \zeta_\Delta \sim \mathcal{N}(0, V_{q,j}^\Delta), \qquad V_{q,j}^\Delta = \frac{\zeta_\Delta^2 (\tau_q^\Delta)^2 (\lambda_{q,j}^\Delta)^2}{\zeta_\Delta^2 + (\tau_q^\Delta)^2 (\lambda_{q,j}^\Delta)^2}. \tag{3.13}$$

This construction separates globally useful feature effects $\boldsymbol{\beta}_{\mathrm{base}}$ from effects that vary across adjacent quantile levels. It is related to interquantile shrinkage and composite-quantile regularization in frequentist quantile regression (Jiang et al., 2014; Zou and Yuan, 2008), but the shrinkage is placed on the Bayesian readout path conditional on the fixed feature design.

The next display rewrites the Q augmented likelihood contributions as one stacked Gaussian block conditional on the exAL latent variables and the quantile-specific intercepts. Let $\boldsymbol{\beta}_{\mathrm{st}} = (\boldsymbol{\beta}_1^\top, \ldots, \boldsymbol{\beta}_Q^\top)^\top$, $\boldsymbol{Z}_{\mathrm{st}} = \boldsymbol{I}_Q \otimes \boldsymbol{Z}$, and $\boldsymbol{\alpha}_{\mathrm{st}} = (\alpha_1 \mathbf{1}_T^\top, \ldots, \alpha_Q \mathbf{1}_T^\top)^\top$. Under the exAL augmentation, define

$$d_{q,t} = \lambda(\gamma_q)\sigma_q s_{q,t} + A(\gamma_q) v_{q,t}, \qquad \omega_{q,t} = B(\gamma_q)\sigma_q v_{q,t},$$

and stack these quantities into $\boldsymbol{d}$ and diagonal $\boldsymbol{\Omega} = \mathrm{diag}(\omega_{q,t})$. The conditional Gaussian likelihood is

$$\boldsymbol{y}_{\mathrm{st}} \mid \boldsymbol{\beta}_{\mathrm{st}}, \boldsymbol{\alpha}, \boldsymbol{d}, \boldsymbol{\Omega} \sim \mathcal{N}(\boldsymbol{\alpha}_{\mathrm{st}} + \boldsymbol{Z}_{\mathrm{st}} \boldsymbol{\beta}_{\mathrm{st}} + \boldsymbol{d}, \, \boldsymbol{\Omega}), \qquad \boldsymbol{y}_{\mathrm{st}} = \mathbf{1}_Q \otimes \boldsymbol{y}.$$

Let $\boldsymbol{H}$ be the block first-difference matrix such that

$$\boldsymbol{H}\boldsymbol{\beta}_{\mathrm{st}} = (\boldsymbol{\beta}_1^\top, (\boldsymbol{\beta}_2 - \boldsymbol{\beta}_1)^\top, \ldots, (\boldsymbol{\beta}_Q - \boldsymbol{\beta}_{Q-1})^\top)^\top.$$

With $\tilde{\boldsymbol{\beta}}_{\mathrm{base}} = (\boldsymbol{\beta}_{\mathrm{base}}^\top, \mathbf{0}_r^\top, \ldots, \mathbf{0}_r^\top)^\top$ and diagonal $\boldsymbol{D}_\Delta$ containing the RHS innovation variances $V_{q,j}^\Delta$, the conditional prior is proportional to

$$\exp\left[-\frac{1}{2}(\boldsymbol{H}\boldsymbol{\beta}_{\mathrm{st}} - \tilde{\boldsymbol{\beta}}_{\mathrm{base}})^\top \boldsymbol{D}_\Delta^{-1} (\boldsymbol{H}\boldsymbol{\beta}_{\mathrm{st}} - \tilde{\boldsymbol{\beta}}_{\mathrm{base}})\right].$$

Consequently, the Gaussian full conditional for the stacked slope path has precision

$$\boldsymbol{K}_\beta = \kappa \boldsymbol{Z}_{\mathrm{st}}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{Z}_{\mathrm{st}} + \boldsymbol{H}^\top \boldsymbol{D}_\Delta^{-1} \boldsymbol{H}, \tag{3.14}$$

and mean

$$\boldsymbol{m}_\beta = \boldsymbol{K}_\beta^{-1} \left[\kappa \boldsymbol{Z}_{\mathrm{st}}^\top \boldsymbol{\Omega}^{-1} (\boldsymbol{y}_{\mathrm{st}} - \boldsymbol{\alpha}_{\mathrm{st}} - \boldsymbol{d}) + \boldsymbol{H}^\top \boldsymbol{D}_\Delta^{-1} \tilde{\boldsymbol{\beta}}_{\mathrm{base}}\right]. \tag{3.15}$$

The sparse, block-banded prior contribution in (3.14) is the central computational object for the joint extension.

The non-crossing interpretation is local in adjacent quantile gaps. For neighboring levels,

$$Q_{p_q}(y_t \mid \mathcal{F}_t) - Q_{p_{q-1}}(y_t \mid \mathcal{F}_t) = (\alpha_q - \alpha_{q-1}) + \tilde{\boldsymbol{x}}_t^\top(\boldsymbol{\beta}_q - \boldsymbol{\beta}_{q-1}).$$

On a bounded feature domain, for example after an explicit scaling or clipping contract that puts each component of $\tilde{\boldsymbol{x}}_t$ in $[-1, 1]$, the inequality $\alpha_q - \alpha_{q-1} \geq \|\boldsymbol{\beta}_q - \boldsymbol{\beta}_{q-1}\|_1$ is sufficient for non-crossing. The adjacent-difference RHS prior does not impose this inequality. Instead, it shrinks $\boldsymbol{\beta}_q - \boldsymbol{\beta}_{q-1}$ toward zero, reducing the size of the feature-dependent term and concentrating the posterior near coherent quantile paths. Exact monotonicity still requires an explicit ordered-intercept parameterization, posterior rejection rule, isotonic projection, or monotone rearrangement. The synthetic joint validation below therefore records raw crossings separately from the declared monotone reporting contract used for scored quantile grids.

3.4 Generalized-Bayes RQR–DESN Interval Readout

The preceding models target one or more conditional quantile ordinates. A separate supplementary readout can instead target a prediction interval directly. For a coverage level $c \in (0, 1)$, the RQR–DESN interval readout uses two exchangeable fixed-feature root surfaces

$$\eta_{1t} = \boldsymbol{x}_t^\top \boldsymbol{\beta}_1, \qquad \eta_{2t} = \boldsymbol{x}_t^\top \boldsymbol{\beta}_2,$$

and reports ordered endpoints

$$L_t = \min(\eta_{1t}, \eta_{2t}), \qquad U_t = \max(\eta_{1t}, \eta_{2t}).$$

The root labels are not lower and upper labels; ordering is applied only when interval endpoints are reported. Let

$$e_t = (y_t - \eta_{1t})(y_t - \eta_{2t}).$$

Then $e_t < 0$ exactly when y_t lies between the two roots. The RQR loss uses this sign information through

$$\rho_c(e) = e\{c - \mathbf{1}(e < 0)\},$$

and the empirical risk is

$$R_{T,c}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_2) = \sum_{t=1}^{T} \rho_c\Big[\{y_t - \boldsymbol{x}_t^\top \boldsymbol{\beta}_1\}\{y_t - \boldsymbol{x}_t^\top \boldsymbol{\beta}_2\}\Big]. \tag{3.16}$$

The Bayesian object used here is a generalized posterior over the two readout coefficient vectors,

$$\Pi_{\mathrm{R}}(d\boldsymbol{\beta}_1, d\boldsymbol{\beta}_2 \mid \boldsymbol{y}, \boldsymbol{X}) \propto \pi_\beta(d\boldsymbol{\beta}_1)\pi_\beta(d\boldsymbol{\beta}_2) \exp\{-\omega_{\mathrm{R}} R_{T,c}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_2)\}, \tag{3.17}$$

where $\omega_{\mathrm{R}} > 0$ is a learning rate (Bissiri et al., 2016). The normalization in (3.17) is over coefficient space, not over future response values. Consequently, RQR–DESN provides posterior summaries for interval endpoints, widths, and midpoints; it is not a response likelihood and does not by itself define posterior predictive response draws.

At the population level, the unrestricted RQR problem has a useful interpretation. Writing $L(x)$ and $U(x)$ for the endpoint functions, under regularity, positive interval width, and an interior solution, the first-order conditions imply

$$\Pr\{L(x) < Y < U(x) \mid X = x\} = c$$

and

$$\mathbb{E}[Y\mathbf{1}\{L(x) < Y < U(x)\} \mid X = x] = c\,\mathbb{E}(Y \mid X = x).$$

Thus the target is a coverage-plus-mean-balance interval (Pouplin et al., 2024). It is not generally an equal-tailed interval, a minimum-width interval, or a highest-density interval. With finite DESN features and regularized linear roots, the fitted readout is best interpreted as an RQR-risk approximation within the selected feature class, with coverage checked empirically in the validation design.

4 Posterior Inference

4.1 General Augmented Posterior Target

The quantile-vector posterior is written so that the individual RHS Q–DESN is the $Q = 1$ collapse of Section 3.3. In that collapse, take α_1 as the single-level intercept, identify $\boldsymbol{\beta}_1$ with the single-level non-intercept slope vector, set $\boldsymbol{\beta}_{\mathrm{base}} \equiv \mathbf{0}_r$, place the RHS prior directly on $\boldsymbol{\Delta}_1 = \boldsymbol{\beta}_1$, and omit all baseline-slope factors. In the notation of Section 3.1, the single-level coefficient vector is then $(\alpha, \boldsymbol{\beta}_1^\top)^\top$, with α playing the role of β_0. This convention lets the same MCMC and VB algorithms serve both the general joint readout and the single-quantile readout.

For the general target, let Θ_Δ collect the local, global, auxiliary, and slab scales for the adjacent-innovation RHS hierarchy. Let Θ_0 collect the corresponding baseline RHS scales, included only when $Q > 1$. Let Θ_Q collect $\boldsymbol{\alpha}$, $\boldsymbol{\beta}_{\mathrm{st}}$, the likelihood parameters, Θ_Δ, and, for $Q > 1$, $(\boldsymbol{\beta}_{\mathrm{base}}, \Theta_0)$. Let $\boldsymbol{a}_Q = \{v_{q,t}, s_{q,t} : q = 1, \ldots, Q;\ t = 1, \ldots, T\}$ collect the exAL auxiliary variables. With the stacked quantities $\boldsymbol{y}_{\mathrm{st}}$, $\boldsymbol{\alpha}_{\mathrm{st}}$, $\boldsymbol{Z}_{\mathrm{st}}$, $\boldsymbol{d}$, and $\boldsymbol{\Omega}$ defined in Section 3.3, the tempered complete-data posterior can be written, up to constants not depending on unknowns, as

$$\begin{aligned} p(\Theta_Q, \boldsymbol{a}_Q \mid \boldsymbol{y}, \boldsymbol{Z}) &\propto \exp\left[-\frac{\kappa}{2}(\boldsymbol{y}_{\mathrm{st}} - \boldsymbol{\alpha}_{\mathrm{st}} - \boldsymbol{Z}_{\mathrm{st}}\boldsymbol{\beta}_{\mathrm{st}} - \boldsymbol{d})^\top \boldsymbol{\Omega}^{-1}(\boldsymbol{y}_{\mathrm{st}} - \boldsymbol{\alpha}_{\mathrm{st}} - \boldsymbol{Z}_{\mathrm{st}}\boldsymbol{\beta}_{\mathrm{st}} - \boldsymbol{d})\right] \\ &\times \prod_{q=1}^{Q}\prod_{t=1}^{T}\left\{(\omega_{q,t})^{-\kappa/2} p(v_{q,t} \mid \sigma_q) p(s_{q,t})\right\} \prod_{q=1}^{Q} p(\sigma_q)\pi_{\gamma_q}(\gamma_q)\mathbf{1}\{L_q < \gamma_q < U_q\} \\ &\times p(\boldsymbol{\alpha})\, p_{\Delta}^{(Q)}(\boldsymbol{\beta}_{\mathrm{st}} \mid \boldsymbol{\beta}_{\mathrm{base}}, \boldsymbol{D}_\Delta)\, p(\Theta_\Delta)\, \mathcal{B}_Q, \end{aligned} \tag{4.1}$$

where

$$\mathcal{B}_Q = \begin{cases} p(\boldsymbol{\beta}_{\text{base}} \mid \Theta_0)p(\Theta_0), & Q > 1, \\ 1, & Q = 1. \end{cases}$$

The Gaussian prior factor $p_{\Delta}^{(Q)}$ is induced by the first-difference matrix $\boldsymbol{H}$ and innovation covariance $\boldsymbol{D}_{\Delta}$ in (3.14)–(3.15). For $Q > 1$, it is the adjacent-innovation prior centered on the baseline slope, so the $\boldsymbol{\beta}_{\text{base}}$ argument is active. For $Q = 1$, there is no random baseline argument, and the same factor is read as the ordinary RHS slope prior

$$p_{\Delta}^{(1)}(\boldsymbol{\beta}_1 \mid \boldsymbol{D}_{\Delta}) \propto \exp\left(-\frac{1}{2}\boldsymbol{\beta}_1^{\top}\boldsymbol{D}_{\Delta}^{-1}\boldsymbol{\beta}_1\right), \qquad \Delta_1 = \boldsymbol{\beta}_1.$$

The AL special case omits $s_{q,t}$, γ_q, and the positive-truncated Normal factors. The ordinary single-quantile posterior is obtained from (4.1) by setting $Q = 1$, $\kappa = 1$, $\boldsymbol{y}_{\text{st}} = \boldsymbol{y}$, $\boldsymbol{Z}_{\text{st}} = \boldsymbol{Z}$, $\boldsymbol{\alpha}_{\text{st}} = \alpha\mathbf{1}_T$, $\boldsymbol{\beta}_{\text{st}} = \boldsymbol{\beta}_1$, and dropping $\mathcal{B}_Q$:

$$p(\alpha, \boldsymbol{\beta}_1, \sigma, \gamma, \{v_t, s_t\}_{t=1}^T, \Theta_{\Delta} \mid \boldsymbol{y}, \boldsymbol{Z}) \propto \exp\left[-\frac{1}{2}(\boldsymbol{y} - \alpha\mathbf{1}_T - \boldsymbol{Z}\boldsymbol{\beta}_1 - \boldsymbol{d}_1)^{\top}\boldsymbol{\Omega}_1^{-1}(\boldsymbol{y} - \alpha\mathbf{1}_T - \boldsymbol{Z}\boldsymbol{\beta}_1 - \boldsymbol{d}_1)\right]$$
$$\times \prod_{t=1}^{T}\left\{(\omega_t)^{-1/2}p(v_t \mid \sigma)p(s_t)\right\}p(\sigma)\pi_{\gamma}(\gamma)\mathbf{1}\{L < \gamma < U\}p(\alpha)\,p_{\Delta}^{(1)}(\boldsymbol{\beta}_1 \mid \boldsymbol{D}_{\Delta})p(\Theta_{\Delta}). \tag{4.2}$$

Thus the posterior, not only the algorithms, has a single general expression with a transparent collapse to the individual RHS Q–DESN. The exAL augmentation preserves the Gaussian slope-path update in (3.14)–(3.15), but the scale-asymmetry blocks (σ_q, γ_q) remain non-conjugate. We therefore use two computation routes for this fixed-design target: an MCMC sampler with separate exAL block updates and a mean-field variational Bayes approximation with Laplace–Delta updates for the non-conjugate blocks. In the algorithms below, GIG, positive-truncated Normal, Gaussian, and inverse-gamma entries denote closed-form full conditionals for MCMC and closed-form coordinate factors for VB. The block order is intentionally the same in both algorithms; only the final exAL scale-asymmetry block requires a non-conjugate update.

4.2 Exact Loss-Kernel Augmentation for RQR–DESN

The RQR–DESN generalized posterior in Section 3.4 has its own augmentation. It is separate from the AL/exAL response augmentations used by Q–DESN. Let

$$r_t = y_t^2, \qquad m_t^{\text{R}} = y_t(\eta_{1t} + \eta_{2t}) - \eta_{1t}\eta_{2t},$$

so that $r_t - m_t^{\text{R}} = e_t$. With

$$\sigma_{\text{R}} = \omega_{\text{R}}^{-1}, \qquad \xi_c = \frac{1 - 2c}{c(1 - c)}, \qquad \phi_c = \frac{2}{c(1 - c)},$$

the exponentiated RQR loss has the exact kernel identity

$$\exp\{-\omega_{\mathrm{R}}\rho_c(e_t)\} = \frac{\sigma_{\mathrm{R}}}{c(1-c)} \int_0^\infty \mathcal{N}\Big(r_t \mid m_t^{\mathrm{R}} + \xi_c v,\ \phi_c \sigma_{\mathrm{R}} v\Big) \frac{1}{\sigma_{\mathrm{R}}} \exp(-v/\sigma_{\mathrm{R}})\, dv. \tag{4.3}$$

This is an exact augmentation of the loss kernel, not a generative model for the original response. The observed y_t appears in both r_t and m_t^{R}, so the pseudo-response form should not be interpreted as a Gaussian observation equation for y_t.

Conditional on the latent scales v_t, each root coefficient block is Gaussian after conditioning on the other root, and the v_t blocks have GIG full conditionals. Ridge and RHS priors enter through the same canonical Gaussian prior-precision form used for Q–DESN readouts, applied separately to the two roots. The complete RQR–DESN MCMC block, including draw-level endpoint ordering, is given in the supplement. Algorithms 1–2 below remain the Q–DESN and quantile-vector Q–DESN algorithms.

4.3 Markov Chain Monte Carlo

The augmented likelihood and shrinkage hierarchy give a partially Gibbs sampler. Closed-form blocks are sampled from their named full conditionals; the complete-data targets, full conditionals, CAVI factors, and ELBO terms for AL and exAL likelihoods under ridge and RHS priors are given in Sections S4–S8 of the supplement. Under the Nishimura–Suchard hierarchy, the global-local scales have inverse-gamma updates. Each exAL block (σ_q, γ_q) remains non-conjugate and is updated with Metropolis–Hastings or slice moves on transformed coordinates. Algorithm 1 targets (4.1); with $Q = 1$, it targets the collapsed posterior in (4.2). Ridge and AL variants omit the corresponding shrinkage, $s_{q,t}$, and γ_q blocks. Diagnostics use trace plots, effective sample size summaries, crossing diagnostics for multi-quantile grids, and monitoring of the (σ_q, γ_q) updates, including acceptance rates when Metropolis–Hastings is used.

Algorithm 1 Block MCMC for the quantile-vector Q–DESN posterior. For $Q = 1$, set $\boldsymbol{\beta}_{\text{base}} = \mathbf{0}_r$, place the RHS prior on $\Delta_1 = \boldsymbol{\beta}_1$, and omit baseline-slope blocks. AL and ridge variants omit the corresponding $s_{q,t}$, γ_q, and shrinkage blocks.

Require: Data $\{y_t, \tilde{\boldsymbol{x}}_t\}_{t=1}^{T}$, quantile grid $p_1 < \cdots < p_Q$, priors, and learning rate κ

Require: MCMC iterations N and convergence-monitoring settings

```
Initialize α, β_st, likelihood parameters, latent variables, and RHS scales
if Q > 1 then
    Initialize β_base and its baseline RHS scales
end if
for i = 1 to N do
    Latents: sample v_{q,t} from independent GIG full conditionals
    Latents (exAL): when exAL is used, sample s_{q,t} from independent positive-truncated
    Normal full conditionals
    Slope path: form K_β and m_β from (3.14)–(3.15)
    Slope path: sample β_st | · ~ N(m_β, K_β^{-1})
    if Q > 1 then
        Baseline: sample β_base from its Gaussian conditional
        Baseline scales: update baseline RHS scales from inverse-gamma full conditionals
    end if
    Intercepts: sample α from its Gaussian conditional, or from an ordered-gap block when
    exact intercept monotonicity is imposed
    RHS scales: sample innovation local, global, auxiliary, and slab scales from inverse-gamma
    full conditionals
    RHS scales (Q = 1): use the ordinary RHS scale updates for β_1
    Likelihood: update each exAL (σ_q, γ_q) block by MH or slice sampling on transformed
    coordinates
    Likelihood (AL): use the corresponding σ_q block and omit γ_q
    Monitor: store post-burn-in draws, crossing diagnostics, and convergence summaries, with
    thinning only if used
end for
```

4.4 Variational Bayes

For lower-cost approximate inference, we use a mean-field variational Bayes approximation for the augmented posterior above. It is an approximation to the same fixed-design model, not a different model. The Gaussian, GIG, positive-truncated Normal, and inverse-gamma factors have closed-form coordinate updates. The exAL scale-asymmetry blocks are non-conjugate. Building on the variational strategy of Barata et al. (2022) and the non-conjugate variational ideas of Wang and Blei (2013), the VB–LD implementation uses independent Laplace approximations for these blocks and Delta-method approximations for expectations of smooth functions of (σ_q, γ_q). This replaces an importance-sampling step with deterministic approximations inside the coordinate updates.

For the quantile-vector QVP–RHS formulation, the mean-field family contains Gaussian factors for $\boldsymbol{\beta}_{\mathrm{st}}$, $\boldsymbol{\alpha}$, and, when $Q > 1$, $\boldsymbol{\beta}_{\mathrm{base}}$; GIG factors for the $v_{q,t}$; positive-truncated Normal factors for the $s_{q,t}$; inverse-gamma factors for baseline and innovation RHS scales; and independent Laplace–Delta factors for the (σ_q, γ_q) blocks. Under the $Q = 1$ collapse, the baseline factor and its scale factors are removed, and the innovation scale factors are exactly the ordinary RHS factors for the single slope readout. Posterior means and predictive summaries from VB–LD are therefore variational approximations, with additional approximation error only from the Laplace–Delta scale-asymmetry blocks.

The Laplace–Delta step uses unconstrained coordinates $\boldsymbol{\eta}_q = (\log \sigma_q, \log\{(\gamma_q - L_q)/(U_q - \gamma_q)\})^\top$, on which the expected complete-data log posterior, including the Jacobian, is approximated by a Normal density at its mode. If $\hat{\boldsymbol{\eta}}_q$ is the mode and Σ_q is the negative inverse Hessian at that mode, then smooth expectations needed by the closed-form updates are approximated by

$$\mathbb{E}\{g(\boldsymbol{\eta}_q)\} \approx g(\hat{\boldsymbol{\eta}}_q) + \frac{1}{2}\mathrm{Tr}\Big\{\nabla^2 g(\hat{\boldsymbol{\eta}}_q)\Sigma_q\Big\}\,.$$

Thus, only expectations involving the exAL scale and asymmetry use Laplace–Delta; the remaining factors are updated in closed form. The updates are iterated until the evidence lower bound (ELBO), or its Laplace–Delta approximation, stabilizes.

Algorithm 2 outlines the coordinate ascent updates for the variational approximation to (4.1); with $Q = 1$, the same updates approximate (4.2). Ridge and AL variants omit the corresponding blocks. The output is a set of approximate posterior moments, predictive summaries, and crossing diagnostics at the target quantile levels.

Algorithm 2 Block VB–LD for the quantile-vector Q–DESN posterior. For $Q = 1$, set $\boldsymbol{\beta}_{\text{base}} = \mathbf{0}_r$, place the RHS prior on $\Delta_1 = \boldsymbol{\beta}_1$, and omit baseline-slope factors. AL and ridge variants omit the corresponding latent, asymmetry, and shrinkage factors.

Require: Data $\{y_t, \tilde{\boldsymbol{x}}_t\}_{t=1}^{T}$, quantile grid $p_1 < \cdots < p_Q$, priors, and learning rate κ
Require: Convergence controls for the ELBO or Laplace–Delta ELBO approximation

```
Initialize factors for q_β(β_st), q_α(α), latent variables, RHS scales, and q_{σ_q,γ_q}(σ_q, γ_q)
if Q > 1 then
    Initialize q_{β_base}(β_base) and its baseline RHS scale factors
end if
repeat
    Latents: update q(v_{q,t}) as independent GIG factors
    Latents (exAL): when exAL is used, update q(s_{q,t}) as independent positive-truncated
      Normal factors
    Slope path: update q_β(β_st) as a multivariate Normal factor
    Slope path: use the expected version of (3.14)
    if Q > 1 then
        Baseline: update q_{β_base}(β_base) as a Gaussian factor
        Baseline scales: update baseline RHS scale factors as inverse-gamma factors
    end if
    Intercepts: update q_α(α) as a Gaussian factor, or use ordered-gap factors when exact
      intercept monotonicity is imposed
    RHS scales: update innovation local, global, auxiliary, and slab scale factors as inverse-
      gamma factors
    RHS scales (Q = 1): use the ordinary RHS factors for β_1
    Likelihood: update each q_{σ_q,γ_q} by a Laplace approximation on unconstrained coordinates
    Likelihood (AL): use the corresponding q_{σ_q} factor
    Expectations: update Delta-method expectations for smooth functions of (σ_q, γ_q)
    Monitor: evaluate the ELBO or Laplace–Delta ELBO approximation, crossing diagnostics,
      and convergence criteria
until convergence
```

4.5 Supplementary Algebra and Implementation Notes

Equations (4.1) and (4.2) define the article-facing posterior targets, and Algorithms 1–2 are placed immediately after them because they implement those targets. The supplement keeps the single-level full conditionals and the full quantile-vector posterior algebra explicit: Sections S4–S7 give the single-level AL/exAL updates, and Section S8 gives the QVP–RHS posterior, stacked Gaussian blocks, VB factorization, ELBO terms, and crossing diagnostics. Thus a reader implementing only one quantile can use Algorithms 1–2 with $Q = 1$ and the collapsed RHS prior, while a reader implementing multiple levels uses the same algorithms with the baseline and adjacent-innovation blocks active.

5 Posterior Prediction and Quantile Synthesis

Prediction and multi-quantile reporting involve separate operations. The first operation propagates posterior or variational uncertainty through the fixed reservoir recursion at future horizons. The second, used only when several levels are reported together, converts fitted level-specific summaries into a monotone curve or instead uses the joint quantile-vector readout introduced in Section 3.3. The subsections below keep these operations separate.

5.1 Multi-Step Posterior Quantile Prediction

Forecasting is conditional on the fitted readout, the fixed reservoir construction, and the working likelihood. At a forecast origin T, the historical reservoir state is obtained by teacher forcing: the recursion in Eq. (2.1) is run through the observed record with observed responses in the lag vector $\boldsymbol{u}_t$. After T, future responses are unobserved, so multi-step forecasting is draw-specific. For a single-level fit at target p_0, write the draw as $\theta_{p_0}^{(m)} = (\boldsymbol{\beta}_{p_0}^{(m)}, \sigma_{p_0}^{(m)}, \gamma_{p_0}^{(m)})$. Conditional on a fixed future design row and this draw, the p_0-indexed working-likelihood predictive distribution is available in closed form:

$$y_{T+h}^{(m,p_0)} \sim \mathrm{exAL}_{p_0}\left(\boldsymbol{x}_{T+h}^{(m,p_0)\top} \boldsymbol{\beta}_{p_0}^{(m)}, \sigma_{p_0}^{(m)}, \gamma_{p_0}^{(m)}\right).$$

The posterior predictive distribution at this fitted level is the mixture of these closed-form working-likelihood distributions over the MCMC posterior draws, or over draws from the VB–LD approximation. It is generally summarized by Monte Carlo rather than by a closed-form mixture. Multi-step horizons add one further Monte-Carlo step: future lagged responses inside the forecast window are generated pathwise, so the future design rows are draw-specific and inherit the same p_0 indexing.

Algorithm 3 Block posterior prediction for one fitted Q–DESN target level p_0.

Require: Target level p_0, forecast origin T, horizon H, and stored reservoir state
Require: Posterior or VB–LD draws for the fitted p_0 model and origin-available future covariates

Condition: teacher-force Eq. (2.1) through time T using observed responses and fixed reservoir weights
for each posterior or variational draw m **do**
  Initialize: set the reservoir state and response lags at the forecast origin
  for $h = 1$ to H **do**
    Design: form $\boldsymbol{u}_{T+h}^{(m,p_0)}$ from known covariates, observed pre-origin lags, and pathwise simulated forecast-window lags
    Reservoir: update Eq. (2.1) and form the draw-specific design vector $\boldsymbol{x}_{T+h}^{(m,p_0)}$
    Readout: compute $\mu_{p_0,T+h}^{(m)} = \boldsymbol{x}_{T+h}^{(m,p_0)\top} \boldsymbol{\beta}_{p_0}^{(m)}$
    Emit: sample from exAL_{p_0} when path propagation or working-likelihood simulation is required
    Report: retain $\mu_{p_0,T+h}^{(m)}$ for p_0-quantile reporting
  end for
end for
Aggregate: summarize draws into forecast bands, p_0-level quantile summaries, scores, and coverage diagnostics

Exogenous covariates must be known at the forecast origin, supplied as scenarios, or generated by an external covariate model. Additional details are given in Section S9 of the supplement. Thus, under independent single-level fits, "forecasting all levels" means applying Algorithm 3 separately to each fitted p_k. A single p_0 fit should not be read as producing fitted target-level quantile summaries for all levels; off-target quantiles implied by its exAL working distribution are not the fitted quantile targets used in Section 5.2.

For the joint quantile-vector readout, the reservoir recursion is unchanged. A posterior or variational draw contains the level-specific readouts $\{\alpha_q, \boldsymbol{\beta}_q, \sigma_q, \gamma_q\}_{q=1}^{Q}$, so the same draw-specific future design vector returns a raw quantile vector over the grid $p_1, \ldots, p_Q$. That raw vector is the joint model output before any monotone reporting contract is applied. The composite quantile-vector likelihood should therefore be read as a device for estimating the conditional quantile path; it does not create a separate replicated future response for each quantile level. In the joint case, prediction over the grid uses the same condition, design, reservoir, and aggregate blocks as Algorithm 3, but replaces the single-level readout step by

$$\mu_{q,T+h}^{(m)} = \alpha_q^{(m)} + \tilde{\boldsymbol{x}}_{T+h}^{(m)\top} \boldsymbol{\beta}_q^{(m)}, \qquad q = 1, \ldots, Q,$$

and reports the resulting raw quantile vector, with optional monotone post-processing.

If recursive lag inputs require a scalar future response path, the joint model uses posterior quantile-function sampling rather than draws from the composite working likelihood. Let $\boldsymbol{q}_{T+h}^{(m)} =$

$(\mu^{(m)}_{1,T+h}, \ldots, \mu^{(m)}_{Q,T+h})^\top$, let $\mathcal{M}$ denote the declared monotone reporting contract, and set $\boldsymbol{q}^{*,(m)}_{T+h} = \mathcal{M}(\boldsymbol{q}^{(m)}_{T+h})$. Interpolating $\{(p_q, q^{*,(m)}_{q,T+h})\}_{q=1}^{Q}$ as in Eq. (5.2) gives a draw-specific quantile function $Q^{*,(m)}_{T+h}(u)$. A synthesized response draw is then

$$u^{(m,r)}_{T+h} \sim \mathrm{Unif}(0,1), \qquad y^{(m,r)}_{T+h} = Q^{*,(m)}_{T+h}\left(u^{(m,r)}_{T+h}\right).$$

Thus the joint readout supplies the quantile function used by the propagation rule; it should not be interpreted as drawing one future response per q, nor as mixing or normalizing the quantile-specific AL/exAL working likelihoods.

5.2 Monotone Quantile Synthesis

The Q–DESN likelihood is specified at one quantile level at a time. When models are fit independently on a grid $p_1 < \cdots < p_K$, their fitted values are target-level quantile summaries, not draws from a joint posterior over a quantile function. Monotone synthesis is useful only when a monotone predictive quantile curve is needed for interval construction, display, or distributional forecast evaluation. The same construction applies to any quantile forecasting method that returns ordered-level summaries; here it remains an optional post-processing step rather than a component of the fitted Q–DESN likelihood.

For each level p_k, let $\hat{q}_{k,t}$ denote a posterior or variational summary of the target quantile at time t, such as the posterior mean or median of $\mu_{k,t} = \boldsymbol{x}_t^\top \boldsymbol{\beta}_k$, or the corresponding forecast-horizon summary from the recursion in Section 5.1; at a forecast origin, t may be read as the target date $T+h$. The synthesis uses the target grid $\{(p_k, \hat{q}_{k,t})\}_{k=1}^{K}$, not off-target quantiles implied by the exAL_{p_k} working likelihood. Since independent fits can cross, first project the grid onto the monotone cone,

$$\boldsymbol{q}^*_t = \operatorname*{arg\,min}_{q_1 \le \cdots \le q_K} \sum_{k=1}^{K} \left(\hat{q}_{k,t} - q_k\right)^2, \tag{5.1}$$

using isotonic regression (Barlow and Brunk, 1972). With $\boldsymbol{q}^*_t = (\hat{q}^*_{1,t}, \ldots, \hat{q}^*_{K,t})$, define the estimated predictive quantile function by piecewise linear interpolation:

$$Q_t^{\mathrm{synth}}(u) = \begin{cases} \hat{q}^*_{1,t}, & u \le p_1, \\ (1-w)\hat{q}^*_{k,t} + w\hat{q}^*_{k+1,t}, & p_k \le u < p_{k+1}, \\ \hat{q}^*_{K,t}, & u \ge p_K, \end{cases} \tag{5.2}$$

where $k = \max\{j : p_j \le u\}$ and $w = (u - p_k)/(p_{k+1} - p_k)$ in the middle line. Endpoint behavior is intentionally conservative: outside the fitted quantile range, the grid does not identify tail shape. Grid CRPS and weighted quantile scores computed from this object are finite-grid approximations to quantile-score integrals under the declared reporting contract, not scores for an unspecified density obtained by mixing or normalizing composite-likelihood factors. If draw-level curves or smoother interpolants are used and residual non-monotonicity appears on a dense evaluation grid, a final monotone rearrangement can be applied before interpolation (Chernozhukov et al., 2010). Independent synthesis and the joint quantile-vector readout should not be interpreted interchangeably. Independent fits preserve the single-target posterior interpretation at each p_k and use synthesis only to report a monotone curve. The joint readout instead couples adjacent quantile-specific slopes during inference; the same monotone contract can be applied afterward only when exact monotone output is required.

6 Model Diagnostics and Reservoir Specification

Prediction and synthesis inherit the reservoir feature map selected before posterior fitting. Reservoir tuning, numerical diagnosis, and forecast evaluation therefore have distinct roles. Reservoir tuning chooses a reproducible feature-generation rule before the final analysis; numerical diagnostics screen candidate feature maps for alignment, stability, redundancy, and conditioning failures; forecast evaluation is defined by the simulation or application protocol. Because posterior inference is conditional on the fixed feature map, this design record is part of the empirical interpretation rather than a post hoc implementation detail. The simulation and streamflow sections specify the concrete fit-recovery and, where applicable, forecast-scoring criteria used in their empirical designs.

Let

$$\mathcal{R} = \{D, \{n_d\}, \{\tilde{n}_d\}, m, T_0, \{\alpha_d\}, \{\rho_d\}, \{\pi_d^{(W)}\}, \{\pi_d^{(\mathrm{in})}\}, f, k, \mathrm{seed}\}$$

denote the depth, layer sizes, reduction sizes, lag memory, washout, leak rates, spectral radii, sparsity probabilities, activations, and random seed used to construct Eq. (2.1). Once $\mathcal{R}$ and the preprocessing convention are fixed, posterior inference is conditional on the resulting design matrix $\boldsymbol{X}$. Specification selection chooses among candidate feature maps before the reported analysis; it does not average over reservoir architectures in the posterior target.

The ESN literature treats these choices as task-dependent design parameters. Reservoir size controls feature richness, leak rates and spectral radii jointly affect memory and stability, and input scaling determines how strongly the reservoir is driven into nonlinear regimes (Jaeger et al., 2007; Lukoševičius, 2012). For deep reservoirs, stacking layers can create hierarchical temporal representations, but additional depth must be justified by validation performance and stability diagnostics rather than by architecture alone (Gallicchio et al., 2018). The condition $\rho_d < 1$ is a useful conservative design choice, but it is not by itself a complete empirical check of the echo state property for a driven nonlinear reservoir (Yildiz et al., 2012). Several ESN modification papers identify failure modes that are visible before forecast scoring: output weights can reveal task-relevant reservoir units, state feedback changes the effective recurrent dynamics through an external feedback channel, redundant state trajectories can damage readout conditioning, and random reservoir realizations can induce substantial seed-to-seed variability (Fan et al., 2017; Ehlers et al., 2025; Saadat et al., 2025; Wu et al., 2018). In Q–DESN these results enter as diagnostics, sensitivity analyses, and possible ablations. They do not change the inferential target: after the feature map passes the validation protocol, the reservoir is fixed for posterior inference.

Component	Role in the fixed feature map	Selection rule	Main diagnostic risk
D	Number of reservoir layers and hierarchy of time scales	Search over a small set and retain additional depth only when validation scores improve after seed checks.	Unnecessary depth can add unstable or redundant features.
$\{n_d\}, \{\tilde{n}_d\}$	Reservoir capacity and lower-layer feature dimension	Treat as the main complexity control, with reduction sizes selected as part of the same feature map.	Large designs can overfit or produce poorly conditioned readouts; redundant states can reduce the effective rank of $\boldsymbol{X}$.
m	Explicit lag memory entering the reservoir input	Choose a domain-informed grid and require all lags to be available at the forecast origin.	Excessive lags increase dimension and can leak future information if alignment is not audited.
$\{\alpha_d\}, \{\rho_d\}$	Reservoir time scale, persistence, and stability	Co-tune because both affect memory. Near-one radii require empirical stability checks.	Saturation, state collapse, or sensitivity to initial conditions.
$\{\pi_d^{(W)}\}, \{\pi_d^{(\mathrm{in})}\}$ and input scaling	Internal connectivity and input excitation	Fix from a small set of literature-guided defaults; broaden the search when diagnostics show under-excitation or saturation.	Too sparse can reduce richness; too dense or strongly driven can saturate nonlinear states.
f, k, T_0	Nonlinearity, bridge transformation, and initial-condition washout	Fix before reservoir selection, with T_0 chosen relative to the memory scale.	Short washout can leave initial-condition artifacts.
Seed	Realization of the random feature map	Treat as a robustness variable, not as a hidden tuning parameter. Refit top candidates over multiple seeds.	A single selected seed can overstate validation performance.

Table 2: Reservoir specification components for the fixed Q–DESN feature map. The selected object is the design used by the Bayesian readout, so selection is based on validation evidence and design diagnostics rather than posterior uncertainty over reservoir weights.

The protocol begins by fixing the empirical design before candidate reservoirs are compared: training and validation windows, held-out test windows, forecast origins, forecast horizons, quantile levels, input transformations, and scoring rules. Each candidate then passes a design check that verifies input alignment, feature dimensions, washout handling, finite state values, empirical spectral radii, state saturation, and readout conditioning. State-matrix diagnostics include near-zero-variance columns, pairwise state correlations, the singular-value or covariance spectrum, effective rank, and a condition number or eigenvalue-spread summary. These summaries distinguish a large reservoir that adds nonredundant dynamics from one that mainly adds collinear features. Candidates that fail the check are excluded before score ranking or sent to a remedial path, such as stronger shrinkage, dimension reduction, or state pruning.

Candidate ranking should combine design diagnostics with validation scores, but the relevant scores depend on the empirical setting. In simulations with a known oracle quantile path, fit-recovery metrics such as quantile-path RMSE, mean absolute error, bias, and correlation are available. In both simulations and applications, forecast scoring must use common forecast origins and horizons for all candidates. For a single fitted quantile level, check loss is the proper score aligned with the target functional. When several levels are fit and an estimated predictive quantile function is reported, interval score, empirical coverage, interval length, CRPS from the synthesized grid, and PIT summaries may also be used, provided the fitted grid is dense enough for the summary being claimed. These scores evaluate the reported single-level working-likelihood forecast or synthesized quantile grid, as applicable; they do not by themselves establish calibrated posterior credible bands under likelihood misspecification.

The coarse search focuses on the primary design parameters in Table 2: D, $\{n_d\}$, $\{\tilde{n}_d\}$, m, $\{\alpha_d\}$, and $\{\rho_d\}$. Secondary choices such as recurrent sparsity, input sparsity, input scaling, and activations are broadened only when diagnostics indicate a specific failure mode. Top candidates are refit over multiple reservoir seeds and, when computationally feasible, compared under both VB–LD and MCMC on a smaller diagnostic subset. The final reservoir specification is frozen before evaluation on held-out test origins. If reservoir ensembles are used, they aggregate only reservoirs that passed the same design checks and are reported as an ensemble sensitivity analysis rather than as posterior uncertainty conditional on a single fixed feature map.

Reproducible reporting includes the candidate grid, fixed defaults, state-quality summaries, rejected candidates and rejection reasons, validation origins, forecast horizons, scoring metrics, reservoir seeds, package and code versions, and the final selected $\mathcal{R}$. Claims about a selected architecture are conditional on this validation design. A single seed, a single forecast origin, or one favorable metric is not evidence of global optimality for the reservoir architecture. Reservoir-adaptation mechanisms such as state feedback or principal-neuron reinforcement are useful future extensions, but they change the feature-generation rule and require their own validation protocol before being combined with the Q–DESN readout.

7 Simulation Validation Study

The simulation evidence is organized around two complementary questions. The first asks whether a single-quantile Q–DESN readout can recover and forecast a known dynamic quantile path under a 500-observation benchmark following the family-based comparison strategy of Yan et al. (2025). The target quantile level is fixed in advance, the data-generating mechanism is known, and competing models are compared under matched error families, fit-window sizes, and quantile levels. The second asks whether a joint multi-quantile readout can recover several known conditional quantile paths under posterior simulation and maintain coherent raw quantile grids over a fixed set of levels. Companion VB–LD screening and forecast-comparison tables are kept in the supplement rather than used as the main article evidence.

In contrast to the linear-regression setting of Yan et al. (2025), the oracle quantile paths are dynamic, and Q–DESN represents those paths through high-dimensional deterministic reservoir features. The design therefore separates two questions: how accurately a fitted model recovers the known conditional quantile path, and how well it forecasts held-out dynamic blocks. We write p for the single target level in the first study and τ for levels in the multi-quantile grid.

The reported main-text tables are MCMC summaries derived from verified source files. The table builders check source hashes, rolling-origin metadata, fit and forecast windows, and table hashes before writing manuscript assets. Run-level provenance remains in the generated manifests, and the supplement preserves the VB–LD companion panels and generated forecast-screening summaries. Larger-window MCMC studies and alternative simulation roots remain outside the present article tables and require separate pinned validation interfaces. The joint multi-quantile validation below has its own registry, table assets, and crossing-diagnostic contract.

7.1 Single-Quantile Dynamic Data-Generating Processes

For each error family f and target quantile level $p \in \{0.05, 0.25, 0.50\}$, the synthetic series is generated as

$$y_{t,p,f} = \mu_{t,f} + \varepsilon_{t,p,f}, \qquad \varepsilon_{t,p,f} = e_{t,f} - H_f^{-1}(p),$$

where H_f is the distribution function of the raw error $e_{t,f}$. The shift by $H_f^{-1}(p)$ makes the p-quantile of $\varepsilon_{t,p,f}$ equal to zero. Hence

$$Q_p(y_{t,p,f} \mid \boldsymbol{\theta}_{t,f}) = \mu_{t,f}.$$

The oracle quantile path is generated by a six-dimensional dynamic linear state with local level, local slope, and two harmonic pairs:

$$\mu_{t,f} = \boldsymbol{F}^\top \boldsymbol{\theta}_{t,f}, \qquad \boldsymbol{F} = (1, 0, 1, 0, 1, 0)^\top,$$

with local linear trend and two sine–cosine harmonic pairs. The seasonal period is 90, the harmonics are 1 and 2, and the state innovation covariance $\boldsymbol{W}_\theta$ has diagonal square roots

$$\sqrt{\operatorname{diag}(\boldsymbol{W}_\theta)} = (0.005,\ 0.00002,\ 0.004,\ 0.004,\ 0.003,\ 0.003).$$

The initial covariance is $\boldsymbol{C}_0 = 0.01\boldsymbol{I}_6$, and each simulated trajectory is initialized deterministically at the family-specific state in Table 3. For a fixed family, the same latent path $\{\mu_{t,f}\}$ is used across the three values of p; only the centering constant $H_f^{-1}(p)$ changes.

Family	Raw error distribution H_f	Initial level	Initial slope	Seasonal settings
Gaussian	$\mathcal{N}(0, 10^2)$	40	0.012	harmonics $(24, 0.35)$, $(8, -0.80)$
Laplace	Laplace with scale 10	35	0.011	harmonics $(28, -0.15)$, $(10, 0.75)$
Gaussian mixture	$0.1\mathcal{N}(0, 0.5^2) + 0.9\mathcal{N}(1, 15^2)$	45	0.014	harmonics $(32, 0.85)$, $(12, -1.25)$

Table 3: Single-quantile dynamic data-generating mechanisms. Raw errors are shifted by $H_f^{-1}(p)$ so that the p-quantile of the observation error is zero; the same latent dynamic quantile path is then shared across target levels within each family. Seasonal entries report amplitude and phase for the two harmonic pairs in the initial state.

7.2 Single-Quantile Fit and Forecast Protocol

Each source trajectory contains $T_{\text{warm}} = 2000$ warmup values and $T_{\text{main}} = 10000$ main source values. All indices below refer to the main source scale after warmup removal. The fitting origin is source index 9000, and the held-out block is 9001:10000. The two fit-window sizes are $T_{\text{fit}} = 500$ and $T_{\text{fit}} = 5000$, corresponding to training windows 8501:9000 and 4001:9000. Forecast evaluation uses a rolling-origin, no-refit state-update protocol over the held-out block. The primary protocol has maximum lead $H_{\max} = 30$ and origin stride 30. The article-facing tables are tied to the verified protocol fields, source hashes, and row manifests exported by the validation study.

7.3 Single-Quantile Competing Methods

For each family and target quantile level in the single-quantile validation bundle, the Q–DESN comparison varies the working likelihood, readout prior, and inference method specified by the validation manifests. To keep table labels compact, entries labeled QDESN use the AL working likelihood and entries labeled exQDESN use the exAL working likelihood; the suffix gives the readout prior, either ridge or regularized horseshoe (RHS). The reservoir specification is fixed within this grid and is recorded through the validation provenance, including the reservoir profile, random seed, source hashes, package version, branch, commit, and run tag. Dynamic quantile linear model (DQLM) and extended dynamic quantile linear model (exDQLM) baselines are fit to the same family, quantile level, fit-window, and forecast-origin protocol.

The primary comparison does not use repeated data-generating roots, a broad reservoir hyperparameter search, non-Bayesian ESN readouts, or post hoc multi-quantile synthesis. Those omissions are deliberate: the goal is to test whether Q–DESN readouts can be competitive with structured dynamic quantile linear models under linear dynamic mechanisms that favor the structured baselines.

7.4 Single-Quantile Criteria for Comparison

Let $\widehat{q}_t$ denote a posterior or variational summary of the fitted conditional p-quantile, such as the posterior median of the readout location. Because the oracle path μ_t is known, fit-window recovery can be evaluated using quantile mean absolute error, quantile RMSE, bias, correlation, and check loss at the target level p. Forecast accuracy is evaluated only on outcomes not used for fitting, under common forecast origins and leads for every model. The article-facing tables report fit recovery and forecast scoring; runtime, row-completion, and source-hash diagnostics remain in the guarded interface files and generated manifest.

For single-level quantile fits, the primary proper score is check loss at the fitted level. Quantile MAE and quantile RMSE compare the fitted or forecast quantile summary directly with the known oracle quantile path, while check loss scores realized observations relative to the reported quantile. These metrics therefore answer different questions and should not be collapsed into one rank. Runtime is retained as a computational criterion for comparing posterior simulation with VB–LD in the generated provenance and supplement, not as a statistical accuracy measure.

Because the simulation uses one reproducible dynamic root for each family and quantile level, the tables report deterministic criterion values for that source design rather than Monte Carlo means over repeated data sets. Any boldface or ranking in the displayed MCMC tables is interpreted within a fixed family, quantile level, fit-window, model family, and metric. A model row is eligible for interpretation only if the shared interface reports successful completion, valid source hashes, valid fit and forecast windows, nonmissing fit-recovery metrics, nonmissing rolling-origin forecast metrics, and an acceptable health or signoff status.

7.5 Single-Quantile Fit and Forecast Results

The verified protocol uses the 500-observation training window 8501:9000, the held-out block 9001:10000, rolling-origin forecasts with leads 1–30 and origin stride 30, and the three target levels $p = 0.05, 0.25, 0.50$. Tables 4–6 report the MCMC family-specific summaries. The entries are generated from lead-level rolling-origin interface rows: fit RMSE measures recovery of the known oracle quantile path on the training window, forecast MAE measures oracle-quantile forecast error averaged across scored rolling-origin lead-target pairs, and check loss scores realized observations at the fitted quantile level. Runtime and row-level provenance are retained in the generated manifest. Boldface marks the lowest MCMC value within each quantile level and metric, not a global ranking across different statistical criteria.

The result tables are derived from the tracked final-validation config and table-builder outputs without changing the generated source tables. The generated manifest records the full validation paths, interface hashes, source-registry hash, table hashes, row counts, audited compact replacements, and any remaining chain-autocorrelation qualifications. The displayed MCMC tables should therefore be interpreted as the article-facing validation bundle for this fixed dynamic design, not as a claim about larger-window MCMC runs, alternative fit sizes, multiple dynamic roots, or an unreported

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss
DQLM	3.036	3.829	1.103	3.811	**2.209**	**3.328**	2.590	**1.161**	**4.029**
exDQLM	**1.935**	16.23	4.313	2.344	6.634	4.132	2.808	1.637	4.109
QDESN ridge	4.836	11.04	2.461	3.839	26.11	8.263	3.258	27.75	14.07
exQDESN ridge	3.798	16.60	1.652	3.367	27.05	8.496	3.259	27.66	14.03
QDESN RHS	3.217	7.930	1.252	2.746	3.310	3.396	1.972	3.791	4.339
exQDESN RHS	2.788	**2.652**	**1.076**	**2.131**	3.557	3.437	**1.927**	3.475	4.285

Table 4: MCMC single-quantile fit-and-forecast comparison for the Gaussian simulation family. Forecast entries average scored rolling-origin lead-target pairs over leads 1–30 with origin stride 30. Lower values are better for all displayed metrics, and boldface marks the lowest MCMC value within each quantile level and metric.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss
DQLM	**3.663**	9.791	1.968	2.850	3.520	4.548	1.774	**1.337**	**5.082**
exDQLM	5.677	22.44	5.154	1.710	6.811	5.106	1.774	2.013	5.206
QDESN ridge	9.972	19.77	2.483	4.018	9.954	5.686	2.967	13.41	8.863
exQDESN ridge	7.853	9.678	2.063	3.265	10.93	5.821	2.949	12.43	8.453
QDESN RHS	6.783	12.14	2.103	2.054	**2.474**	**4.493**	**1.217**	2.765	5.254
exQDESN RHS	6.671	**2.143**	**1.859**	**1.564**	3.921	4.608	1.263	3.248	5.334

Table 5: MCMC single-quantile fit-and-forecast comparison for the Laplace simulation family. Forecast entries average scored rolling-origin lead-target pairs over leads 1–30 with origin stride 30. Lower values are better for all displayed metrics, and boldface marks the lowest MCMC value within each quantile level and metric.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss	Fit RMSE	Forecast MAE	Check loss
DQLM	2.623	3.003	**1.505**	4.740	**2.487**	**4.540**	2.226	**1.932**	**5.556**
exDQLM	**2.554**	23.91	5.857	**1.381**	8.941	5.649	2.408	2.207	5.607
QDESN ridge	8.180	8.722	1.930	5.094	16.82	6.924	4.273	8.097	7.012
exQDESN ridge	6.316	4.912	1.619	3.917	13.17	6.236	4.557	7.532	6.780
QDESN RHS	6.269	4.440	1.562	2.659	4.246	4.690	**1.229**	3.141	5.690
exQDESN RHS	4.176	**2.028**	1.510	2.069	5.376	4.803	1.384	3.423	5.727

Table 6: MCMC single-quantile fit-and-forecast comparison for the Gaussian-mixture simulation family. Forecast entries average scored rolling-origin lead-target pairs over leads 1–30 with origin stride 30. Lower values are better for all displayed metrics, and boldface marks the lowest MCMC value within each quantile level and metric.

reservoir hyperparameter search. The VB–LD panels from the same generated validation bundle are reported in the supplement as computational companion evidence.

7.6 Joint Multi-Quantile Frozen-Feature Validation

The second simulation study asks a different question: whether a shared quantile-vector readout can recover several conditional quantile paths at once under a frozen feature design. VB and VB–LD are used here for screening, calibration, and initialization. The article-facing evidence is the balanced VB-initialized MCMC confirmation grid over the four readout–likelihood combinations: Joint QDESN RHS, Independent QDESN RHS, Joint exQDESN RHS, and Independent exQDESN RHS. These rows evaluate posterior quantile-grid readout paths; they should not be read as validation of a scalar predictive density obtained by multiplying or mixing the component working likelihoods.

The selected readout is fit across

$$\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}.$$

The balanced confirmation grid uses eight synthetic data-generating mechanisms, a frozen feature design, and a 500-observation fit window. The three bridge mechanisms align with the Gaussian, Laplace, and Gaussian-mixture families in the single-quantile study. The five stress mechanisms introduce Student-t errors, asymmetric tails, persistent heavy tails, regime shifts, and nonlinear reservoir-friendly dynamics. The resulting grid has 32 scenario–model MCMC rows. Quantile summaries are compared with the known conditional quantiles, realized observations are scored through check loss and grid CRPS, and empirical hit-rate errors summarize marginal calibration. Reported scores use the declared monotone quantile-grid contract; raw adjacent-level crossings before that contract are retained as numerical coherence diagnostics.

Scenario	Mechanism detail	Joint Q–DESN AL–RHS	Independent Q–DESN AL–RHS	Joint exQ–DESN exAL–RHS	Independent exQ–DESN exAL–RHS
Asymmetric Laplace Tail	stress; Asymmetric Laplace; AR(1) seasonal location–scale	**0.084 (0.083; 1)**	0.091 (0.088; 2)	0.094 (0.084; 0)	0.090 (0.078; 0)
Gaussian-Mixture Bridge	bridge; Gaussian mixture; AR(1) seasonal location–scale	0.095 (0.096; 0)	0.101 (0.103; 0)	0.101 (0.103; 0)	**0.095 (0.103; 0)**
Laplace Bridge	bridge; Laplace; AR(1) seasonal location–scale	**0.087 (0.090; 0)**	0.100 (0.092; 18)	0.114 (0.122; 0)	0.113 (0.118; 0)
Nonlinear Reservoir Friendly	stress; Gaussian mixture; nonlinear reservoir-friendly	**0.108 (0.109; 0)**	0.115 (0.113; 8)	0.153 (0.160; 0)	0.154 (0.151; 0)
Normal Bridge	bridge; Gaussian; AR(1) seasonal location–scale	**0.077 (0.081; 0)**	0.082 (0.082; 1)	0.115 (0.102; 0)	0.105 (0.101; 0)
Persistent Heavy Tail	stress; Student-t; AR(1) seasonal location–scale	0.130 (0.118; 0)	0.130 (0.120; 0)	0.121 (0.108; 0)	**0.107 (0.099; 0)**
Regime Shift	stress; Student-t; regime-shift location–scale	0.174 (0.087; 0)	0.091 (0.091; 4)	0.188 (0.093; 0)	**0.090 (0.086; 0)**
Student-t Location-Scale	stress; Student-t; AR(1) seasonal location–scale	**0.068 (0.061; 0)**	0.071 (0.063; 4)	0.100 (0.100; 0)	0.103 (0.098; 0)

Table 7: Scenario-level balanced MCMC confirmation summary for the joint multi-quantile validation study. Entries are forecast MAE with fit-window MAE and raw adjacent-level crossing pairs in parentheses, written as forecast MAE (fit MAE; raw crossings). Boldface marks the lowest forecast MAE within each scenario. Q–DESN denotes the AL working likelihood, exQ–DESN denotes the exAL working likelihood, and RHS denotes the regularized horseshoe prior.

Table 7 summarizes the balanced MCMC confirmation layer at the scenario level. Entries report forecast MAE against the oracle quantile grid, with fit-window MAE and raw adjacent-level crossing counts retained in parentheses; boldface marks the within-scenario forecast-MAE winner. The scenario-wise display makes the heterogeneity clearer than an aggregate model ranking. Joint QDESN RHS is the forecast-MAE winner in five of the eight mechanisms, including the Gaussian, Laplace, asymmetric-tail, nonlinear-reservoir, and Student-t location-scale cases. Independent exQDESN RHS wins the Gaussian-mixture bridge, persistent-heavy-tail, and regime-shift cases. The independent AL readout is competitive in average forecast MAE but carries most of the raw crossing burden, whereas the exAL rows are raw-crossing free in this balanced MCMC packet. The supplement records the protocol audit, aggregate summaries, and metric-specific winners so these qualifications remain reproducible without making the main text table-heavy.

7.7 RQR–DESN Interval-Readout Confirmation

A final simulation-validation layer evaluates the RQR–DESN interval readout defined in Section 3.4 and computed using the loss-kernel augmentation in Section 4.2. This is not another quantile-grid model. The study asks whether direct coverage-indexed interval roots can recover oracle endpoints in fixed designs and improve held-out interval scores on teacher-forced DESN features. The validation is interpreted through ordered endpoints, width, empirical coverage, interval score, and paired baseline deltas; no response-predictive density or recursive response simulation is claimed.

The confirmation reported here follows a broad package-side screen and then reruns the selected MCMC interval-readout configurations with fresh seeds. The targeted denominator has 912 scenario rows: 648 MCMC rows and 264 empirical train-interval baseline rows, with no VB rows. All completion, failure, metric-count, MCMC-diagnostic, interval-validity, and no-response-predictive contract gates pass. The fixed-design stage permits endpoint-recovery interpretation because oracle endpoints are available. The dynamic stage is restricted to teacher-forced held-out interval scoring on fixed DESN features; it is not a recursive response-simulation study.

Table 8 reports the calibration-constrained MCMC winner in each stage, mechanism, and coverage cell. All selected rows improve over the empirical train-interval baseline in paired interval score. Ten of the twelve winner cells have green calibration, and the two yellow cells occur in the nonlinear dynamic DESN mechanism, where the coverage errors are positive but remain within the predeclared review band. These results support RQR–DESN as a promising interval-readout extension of the fixed-feature workflow, while leaving VB calibration, recursive interval forecasting, and nested multi-coverage interval modeling as separate future work.

8 Application: GloFAS Streamflow Forecast Calibration

The empirical application considers medium-range streamflow forecast calibration using the Global Flood Awareness System (GloFAS) and a reference gauge record. GloFAS is a global hydrological forecast and monitoring system jointly developed by the European Commission and the European Centre for Medium-Range Weather Forecasts (ECMWF) and operated within the Copernicus

Stage	Mechanism	c	Selected readout	Interval score	Coverage	Δ score	Cal.
Fixed design	Heavy Tail	0.8	fixed/Ridge/1.5	1.616	0.818	-1.001	Green
Fixed design	Heavy Tail	0.9	fixed/Ridge/1	2.297	0.909	-0.960	Green
Fixed design	Heteroskedastic	0.8	fixed/Ridge/1	1.367	0.761	-1.524	Green
Fixed design	Heteroskedastic	0.9	fixed/Ridge/0.5	1.671	0.944	-2.503	Green
Fixed design	Skewed	0.8	fixed/RHS/1.5	1.113	0.844	-1.128	Green
Fixed design	Skewed	0.9	fixed/RHS/1.5	1.407	0.927	-1.056	Green
Fixed design	Symmetric	0.8	fixed/Ridge/1.5	1.307	0.812	-1.096	Green
Fixed design	Symmetric	0.9	fixed/Ridge/1	1.586	0.916	-1.382	Green
Dynamic DESN	Nonlinear	0.8	DESN/Ridge/1.5	0.752	0.864	-1.643	Yellow
Dynamic DESN	Nonlinear	0.9	DESN/Ridge/1.5	0.941	0.960	-1.646	Yellow
Dynamic DESN	Regime Shift	0.8	DESN/Ridge/1	0.873	0.802	-0.620	Green
Dynamic DESN	Regime Shift	0.9	DESN/Ridge/1.5	1.037	0.925	-0.653	Green

Table 8: RQR–DESN targeted-confirmation winners. The coverage target is c. Selected readouts are reported as design/prior/learning-rate. Δ score is the paired interval-score difference relative to the empirical train-interval baseline, so negative values favor the RQR–DESN readout. Calibration classes use the package-side audit thresholds, with green indicating absolute coverage error at most 0.05 and yellow at most 0.075.

Emergency Management Service (Alfieri et al., 2013; Copernicus Emergency Management Service, 2026). Its operational forecast and reforecast products provide ensemble river-discharge information for flood early warning and forecast verification (Harrigan et al., 2023). Previous calibration work has focused on hydrological model parameters using daily streamflow observations (Hirpa et al., 2018); here Q–DESN is used as a statistical quantile-calibration layer for one forecast system and one audited forecast origin.

8.1 Data and Forecast-Origin Protocol

Let T denote a forecast origin and let H be the issued forecast horizon. The historical reference record is y_t, the transformed USGS streamflow through time T. The retrospective GloFAS product over the same period is g_t^{ret}, and $g_{T+h,m}^{\text{ens}}$ denotes GloFAS ensemble member m issued at origin T for horizon $h = 1, \ldots, H$. The values $y_{T+1}, \ldots, y_{T+H}$ are not available at the forecast origin. They are held out during prediction and used only later for validation.

The selected manuscript-facing run uses the forecast origin 2022-12-25 and scores seven fitted quantile levels over 28 issued target dates. The application is therefore an audited reference case rather than a basin-wide or multi-origin operational evaluation. This restriction is intentional: it keeps the data extraction, forecast-origin alignment, posterior prediction contract, and promotion manifest inspectable before broader comparisons are reported.

8.2 Application Model

For $t \leq T$, the application uses two deterministic Q–DESN feature maps. The reference block $\boldsymbol{x}_t^{\text{ref}}$ is driven by the transformed reference history and origin-available covariates. The discrepancy block $\boldsymbol{x}_t^{\text{disc}}$ is driven by the retrospective discrepancy $g_t^{\text{ret}} - y_t$, using the same audited preprocessing, lag, reservoir, reducer, and washout conventions. For $t > T$, both feature maps are recomputed inside posterior prediction, because output lags may depend on the latent future reference path. For each posterior draw, the forecast-window discrepancy reservoir is driven by the horizon-keyed contrast between the empirical GloFAS ensemble quantile $\hat{q}^{\text{ens}}_{p_0,T+h}$ and the draw-specific future reference path. Thus forecast-window design rows are posterior-predictive quantities, not fixed rows that can be constructed before prediction.

For a fixed quantile level p_0, the calibrated reference quantile is represented by one Q–DESN readout and the GloFAS discrepancy by a second readout,

$$q^{\text{ref}}_{p_0,t} = \boldsymbol{x}_t^{\text{ref}\top}\boldsymbol{\beta}_{p_0}, \qquad d^{\text{glo}}_{p_0,t} = \boldsymbol{x}_t^{\text{disc}\top}\boldsymbol{\alpha}_{p_0}, \qquad q^{\text{glo}}_{p_0,t} = q^{\text{ref}}_{p_0,t} + d^{\text{glo}}_{p_0,t}.$$

The path $q^{\text{ref}}_{p_0,t}$ is the target reference-process quantile, while $d^{\text{glo}}_{p_0,t}$ is the quantile-specific discrepancy between GloFAS and the reference process. The retrospective and ensemble GloFAS products inform the forecast-system quantile path $q^{\text{glo}}_{p_0,t}$; the calibrated forecast target remains $q^{\text{ref}}_{p_0,T+h}$.

At forecast origin T, the issued GloFAS ensemble supplies information about the forecast-system quantile at each covered target date, not a direct forecast of the reference quantile. The Bayesian prediction object is draw-based. For posterior draw $s = 1, \ldots, S$, the draw contains a future reference path, the implied forecast-window reservoir states, and the readout coefficients:

$$q^{\text{ref},(s)}_{p_0,T+h} = \boldsymbol{x}_{T+h}^{\text{ref},(s)\top}\boldsymbol{\beta}^{(s)}_{p_0}, \qquad q^{\text{glo},(s)}_{p_0,T+h} = q^{\text{ref},(s)}_{p_0,T+h} + \boldsymbol{x}_{T+h}^{\text{disc},(s)\top}\boldsymbol{\alpha}^{(s)}_{p_0}.$$

Equivalently, $q^{\text{ref},(s)}_{p_0,T+h} = q^{\text{glo},(s)}_{p_0,T+h} - d^{\text{glo},(s)}_{p_0,T+h}$. Posterior means, medians, intervals, and monotone multi-quantile summaries are computed only after this draw-level construction.

The working-likelihood specification is

$$\begin{aligned}
y_t \mid q^{\text{ref}}_{p_0,t}, \sigma_{\text{ref}}, \gamma_{\text{ref}} &\sim \text{exAL}_{p_0}(q^{\text{ref}}_{p_0,t}, \sigma_{\text{ref}}, \gamma_{\text{ref}}),\\
g^{\text{ret}}_t \mid q^{\text{glo}}_{p_0,t}, \sigma_{\text{glo}}, \gamma_{\text{glo}} &\sim \text{exAL}_{p_0}(q^{\text{glo}}_{p_0,t}, \sigma_{\text{glo}}, \gamma_{\text{glo}}),\\
g^{\text{ens}}_{T+h,m} \mid q^{\text{glo}}_{p_0,T+h}, \sigma_{\text{glo}}, \gamma_{\text{glo}} &\sim \text{exAL}_{p_0}(q^{\text{glo}}_{p_0,T+h}, \sigma_{\text{glo}}, \gamma_{\text{glo}}), \quad h = 1, \ldots, H, \quad m = 1, \ldots, M.
\end{aligned}$$

This display gives the general exAL form. The AL specialization used by the selected reference-case run omits s-latent variables and γ blocks, and replaces the exAL constants by the corresponding AL constants at the fitted level. The issued ensemble members enter as conditionally independent likelihood contributions given the forecast-system quantile path and source-specific likelihood parameters. ECMWF ensemble forecasts use perturbed initial states and model perturbations to produce forecast alternatives (European Centre for Medium-Range Weather Forecasts, 2026); in the Q–DESN calibration layer, this supports treating GloFAS members at a common forecast origin and horizon as exchangeable draws around $q^{\text{glo}}_{p_0,T+h}$. In exAL variants, the default application model shares σ_{glo} and γ_{glo} between retrospective and issued-ensemble GloFAS rows; under the AL specialization, only the corresponding σ_{glo} block is shared.

For this reference-case analysis, the seven fitted quantile levels are 0.05, 0.15, 0.35, 0.50, 0.65, 0.80, and 0.95. They are fitted independently with the AL special case and the VB approximation. This choice is deliberate: the streamflow application is intended as a fast calibration workflow, so the reported run uses Q–DESN under AL rather than exQDESN under exAL. This application therefore uses the independent-fit and synthesis workflow, not the joint quantile-vector readout of Section 3.3. The draw-level predictive summaries are then passed through the monotone synthesis step. The selected manuscript output additionally uses a no-refit spread calibration at the synthesis stage, centered at quantile 0.50, with multiplicative factor 1.4 and additive half-width 0.5. This adjustment changes only the synthesized forecast-window bands; it does not alter the reservoir features, likelihood, priors, or fitted readout coefficients.

The selected fit uses separate DESN feature maps for the reference quantile path and the GloFAS discrepancy path. Both maps have depth $D = 4$, layer widths $(100, 100, 100, 100)$, reducer widths $(100, 100, 100)$, memory 300, washout 500, spectral radii $(0.95, 0.95, 0.95, 0.95)$, recurrent sparsity $(0.03, 0.03, 0.03, 0.03)$, and dense input matrices. The reference map uses leak rates $(0.0225, 0.0225, 0.0225, 0.0225)$ and global/bias input scales 0.16/0.16, and reservoir seed 20260512; the discrepancy map uses leak rates $(0.0075, 0.0075, 0.0075, 0.0075)$, global/bias input scales 0.28/0.28, and reservoir seed 20261521.

The two readout blocks use separate regularized-horseshoe global scales, $\tau_{0,\text{ref}} = 0.001$ and $\tau_{0,\text{disc}} = 0.03$, respectively. The tracked config snapshot and selection manifest record the engine commit, source hashes, selected file hashes, and current manuscript aliases for this run.

8.3 Reference Application Results

Table 9 reports the forecast scores for the selected reference run. Relative to raw GloFAS, the Q–DESN calibration reduces the mean check loss from 0.7639 to 0.3339, the interval score from 13.0538 to 4.1402, and the quantile-grid CRPS from 1.4424 to 0.6859. Mean empirical predictive interval coverage increases from 0.000 to 0.833. These summaries describe one audited forecast origin and should be read as a reference-case calibration result rather than as a multi-origin operational validation.

Table 9: Forecast scoring for the selected GloFAS application run. Scores are computed on the transformed streamflow scale for the audited forecast origin 2022-12-25. Lower check loss, interval score, and CRPS are better; mean coverage reports empirical predictive interval coverage across the scored nominal intervals.

Model	Horizons	Check	Interval	CRPS	Coverage	Reduction
Q–DESN calibration	28	0.3339	4.1402	0.6859	0.833	56.3%
Raw GloFAS	28	0.7639	13.0538	1.4424	0.000	Reference

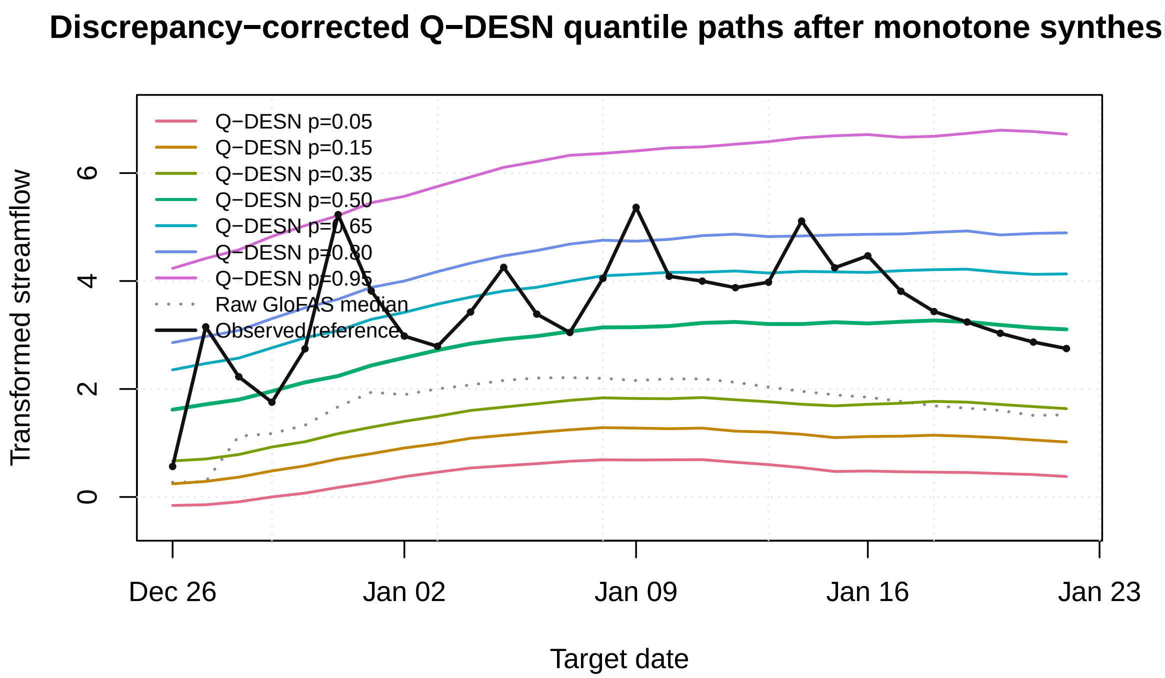

Figure 2: Raw GloFAS quantile paths, discrepancy-corrected Q–DESN quantile paths after monotone synthesis and synthesis-time spread calibration, and held-out reference streamflow over the issued forecast window for the selected GloFAS application run.

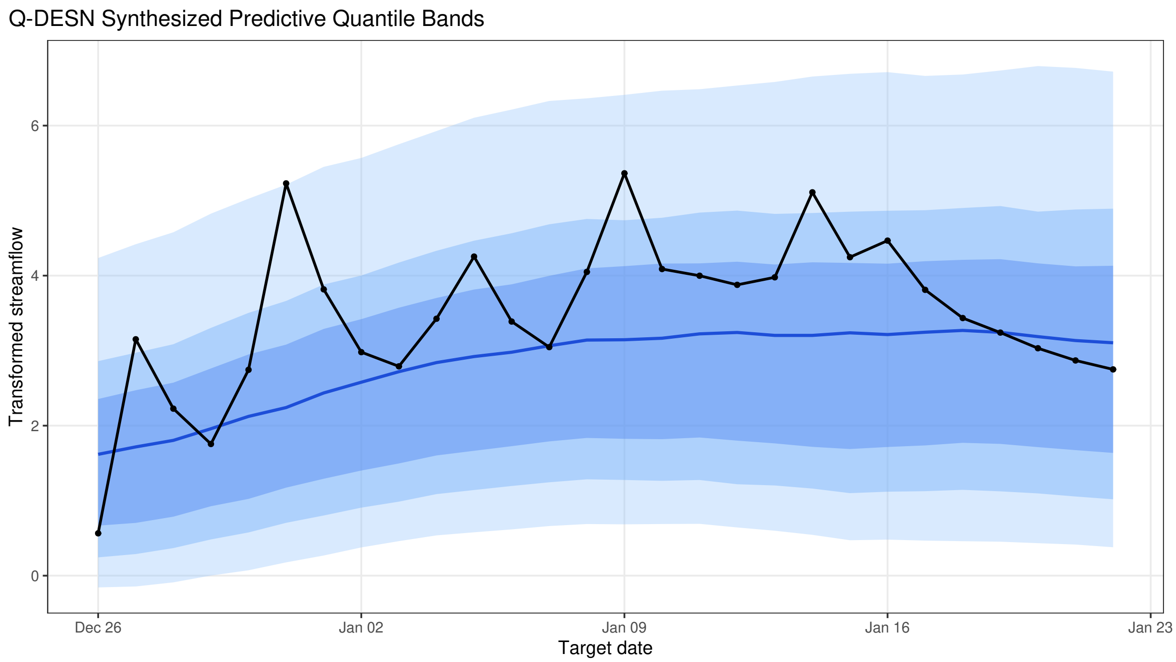

Figure 3: Posterior forecast-window diagnostic for the selected GloFAS application run. The display checks the synthesized and spread-calibrated quantile bands around the forecast origin and the issued target dates used in Table 9.

The selected run passed the source-hash, file-hash, score-table, and VB convergence checks used by the article workflow. The VB fits used median 150; max 150 iterations. Future model specifications can replace this reference case through the same selection manifest and current-output registry, without changing the application model statement.

9 Application: European Electricity Price Forecasting

As a second empirical case, we consider probabilistic day-ahead electricity price forecasting using the PriceFM benchmark of Yu et al. (2026). The article-side target is original-unit quantile scoring on the PriceFM paper quantile grid under fold-aligned test timestamps. The released data cover European bidding regions at 15-minute resolution and include day-ahead price, load, solar, and wind forecasts. PriceFM itself is a graph-informed foundation model: it uses region-level exogenous information and a sparse graph mask derived from the European transmission topology. Our article-side comparison is therefore designed as a scoped benchmark against cached fold-aligned PriceFM predictions, not as a claim about the full paper-wide PriceFM evaluation.

9.1 Comparison Protocol

The article-facing benchmark contains 114 region/fold evaluations, spanning 38 European bidding regions and 3 rolling folds. Each evaluation is scored on the same fold-aligned test timestamps used by the cached PriceFM predictions. The scoring metric is original-unit average quantile loss (AQL) on the PriceFM paper quantile grid $0.10, 0.25, 0.45, 0.50, 0.55, 0.75, 0.90$. Throughout this section, Δ AQL denotes Q–DESN AQL minus PriceFM AQL. Negative values indicate lower Q–DESN AQL; positive values indicate lower cached PriceFM AQL.

The benchmark panel is assembled from audited validation registries and previously computed paper-quantile score files. This assembly step is registry-only: it does not fit new models and it does not alter the cached PriceFM predictions. Across the benchmark, 58 selected region/fold evaluations use target-only information and 56 selected region/fold evaluations use graph-derived PriceFM neighborhood summaries. The benchmark is interpreted as an application of the selected Q–DESN forecasting workflow recorded by the validation registries. It is not used as evidence for the joint quantile-vector RHS prior unless a future PriceFM run explicitly fits and promotes that joint readout under the same fold-aligned contract.

9.2 Full-Surface Results

Across the aligned benchmark, Q–DESN has lower AQL in 66 of 114 region/fold evaluations (57.9%), 12 evaluations are near-equivalent under the monitoring threshold, and cached PriceFM has lower AQL in 36 evaluations. The mean Q–DESN AQL is 6.826, compared with mean PriceFM AQL 7.039, giving mean Δ AQL -0.213 and median Δ AQL -0.083. The aggregate difference is therefore modest rather than uniform. Table 10 shows that the mean Δ AQL is most negative in fold 2, modestly negative in fold 1, and essentially neutral in fold 3.

Table 10: PriceFM benchmark summary. AQL is computed on the original electricity-price scale using the PriceFM paper quantile grid. The last two columns report Q–DESN AQL minus PriceFM AQL; negative values indicate lower Q–DESN AQL and positive values indicate lower cached PriceFM AQL. Near ties are retained as a separate monitoring category.

Scope	Region-fold evaluations	Lower Q–DESN AQL	Near-equivalent AQL	Lower PriceFM AQL	Q–DESN AQL	PriceFM AQL	Mean Δ	Median Δ
Overall	114	66	12	36	6.826	7.039	-0.213	-0.083
Fold 1	38	22	6	10	7.047	7.297	-0.250	-0.073
Fold 2	38	30	2	6	6.429	6.826	-0.397	-0.295
Fold 3	38	14	4	20	7.001	6.993	0.008	0.281

The input set also matters. Graph-derived covariates have the most negative average Δ AQL, whereas target-only evaluations are closer to neutral. This comparison should not be read as a causal attribution to graph features, because input sets and selected model specifications vary across region/fold evaluations. It is a diagnostic summary of where the selected registry is strongest on the aligned benchmark.

Table 11: PriceFM benchmark by input set. Graph-derived inputs include selected evaluations using graph-neighborhood summaries or graph-neighborhood lag features. Target-only inputs use only the target-region information available to the local forecasting workflow.

Input set	Region-fold evaluations	Lower Q–DESN AQL	Near-equivalent AQL	Lower PriceFM AQL	Mean Δ	Median Δ
Graph-derived inputs	56	34	5	17	-0.366	-0.113
Target-only inputs	58	32	7	19	-0.064	-0.046

9.3 Horizon Diagnostics and Remaining Gaps

Horizon-block diagnostics are available for 72 of the 114 region/fold evaluations. These diagnostics preserve the same convention as the aggregate comparison: negative Δ values indicate lower Q–DESN AQL. They should be read as a partial diagnostic layer rather than as a second full-surface registry, because not every audited source artifact retained horizon-block scores.

Table 12: Available horizon-block diagnostics for the PriceFM benchmark. Entries denote scored region/fold and horizon-block combinations for the subset with retained horizon diagnostics. Negative Δ values indicate lower Q–DESN AQL; positive values indicate lower cached PriceFM AQL.

Horizon block	Region-fold horizon blocks	Lower Q–DESN AQL	Mean Δ	Median Δ
1-24	72	22	0.421	0.400
25-48	72	45	-0.683	-0.382
49-72	72	34	-0.196	0.032
73-96	72	42	-0.152	-0.114

The resulting evidence supports a conservative scoped statement. Q–DESN is a competitive reservoir baseline against cached fold-aligned PriceFM predictions on the aligned benchmark panel, with lower mean and median AQL in aggregate. It is not uniformly lower than PriceFM: fold 3 is nearly tied on average, and cached PriceFM has lower AQL for a nontrivial subset of region/fold evaluations. Subsequent read-only repair audits of those PriceFM-favored evaluations did not produce a validation-selected Q–DESN candidate that passed the aligned original-scale test AQL gate, so they are treated as negative diagnostic evidence rather than as a basis for changing the article tables or figures. A new PriceFM run should therefore require a pre-specified change to the design: for example, the information set, model class, selection rule, or joint readout protocol. A fully joint PriceFM run would be needed before making application-level claims about the joint multi-quantile readout.

10 Discussion

Q–DESN treats the fixed DESN reservoir as a nonlinear feature map and places Bayesian quantile inference on the readout. This separation is useful computationally: posterior simulation and VB–LD update only the readout, likelihood, and shrinkage parameters, while the recurrent weights, reducers, scaling constants, and washout convention remain fixed. The empirical results therefore evaluate a conditional Bayesian readout workflow. They should not be interpreted as posterior averaging over reservoir architectures or as a fully Bayesian treatment of recurrent network uncertainty.

The likelihood is also deliberately limited in interpretation. The AL and exAL families are used as quantile-targeting working likelihoods. They provide posterior targets and conditionally tractable augmented updates, but credible bands from the fitted working likelihood are not automatic frequentist predictive coverage guarantees under misspecification. The simulation and application sections therefore report fit recovery, forecast scoring, empirical predictive coverage, runtime, convergence, and provenance diagnostics rather than relying on posterior summaries alone. Calibration of learning rates, sandwich adjustments, or general-Bayes scaling remains an important sensitivity direction, especially for composite multi-quantile likelihoods.

The multi-quantile construction makes the same distinction. Independent single-level Q–DESN fits followed by isotonic regression or rearrangement produce monotone reported quantile curves, but they do not define a joint posterior over quantile-indexed readouts. The joint quantile-vector extension shares information across quantile levels through adjacent RHS slope innovations, giving a soft non-crossing mechanism. It does not impose the non-crossing inequalities as hard constraints, and its composite working likelihood is not a unique response predictive density. When exact monotonicity is required for downstream use, the reported curve should still be obtained through an ordered-intercept parameterization, projection, rearrangement, or another explicitly declared monotone contract.

The validation evidence is correspondingly scoped. The single-quantile dynamic simulations test recovery and held-out forecasting under pinned data-generating mechanisms. The joint multi-quantile synthetic study checks coherence under known conditional quantile paths. The RQR–DESN confirmation evaluates a coverage-indexed interval readout whose ideal population target is coverage-plus-mean balance rather than an equal-tailed or shortest-interval rule; the evidence is through MCMC interval scores, empirical coverage, and paired baseline deltas, not through a response predictive likelihood or recursive simulation study. The GloFAS application is an audited one-origin streamflow calibration case using computationally light independent Q–DESN AL–VB fits and monotone synthesis, while the PriceFM comparison is a fold-aligned benchmark against cached Phase-I predictions. These studies support the reported workflows under their own protocols, not a universal ranking of Q–DESN over all dynamic quantile or foundation-model alternatives.

Several extensions are natural. On the statistical side, future work should study calibrated composite-likelihood scaling, hard or probabilistic non-crossing constraints, richer covariance structures for quantile innovations, and sensitivity to reservoir scaling on bounded feature domains. For RQR–DESN, future work should study calibrated VB uncertainty, learning-rate sensitivity, and joint nested intervals across several coverage levels. On the computational side, the joint model calls for sparse precision solvers, selected covariance extraction for VB, and diagnostics comparing VB–LD with MCMC on small joint problems. On the application side, broader multi-origin GloFAS studies and explicitly joint PriceFM runs are needed before the joint quantile-vector prior can be interpreted as application-level evidence rather than a synthetic validation result.

Within these boundaries, the article establishes Q–DESN as a reproducible fixed-feature Bayesian quantile-readout workflow whose inference, synthesis, diagnostics, and empirical validation are tied to explicit protocols.

Acknowledgments

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